

GENERAL SPECIFICATION G8.2

Finishes for Buildings

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G8.2 FINISHES FOR BUILDINGS

G8.2.1 External and Internal Building Finishes

- (a) This Specification covers external and internal finishes to buildings, and waterproofing.
- (b) References in this Section of the Specification to plaster and plastering shall also be applicable to render and rendering as appropriate.
- (c) Materials for plaster and rendering shall be from approved sources and shall comply with the requirements as follows:
 - (i) Sand : Sand shall be clean, sharp, sand suitable for use in plastering and shall be approved by the S.O.. It shall contain not more than 5% of clay. Under no circumstances shall the approved type of sand be changed during the course of the Works.
 - (ii) Cement : Cement shall be ordinary portland cement, fresh and of an approved brand complying with BS EN 197-1:2011.
 - (iii) Water : Water shall be NEWater or potable water quality, free of vegetation and other deleterious matter.
 - (iv) Waterproofing compound : Waterproofing compound shall be as approved by the S.O. and shall be used strictly in accordance with the manufacturer's recommendations.
- (d) The mortar for lime plastering shall be gauged in the proportion of one (1) part of cement to one (1) part of lime and three (3) parts of sand.
- (e) The mortar for cement rendering shall be gauged in the proportion of one (1) part of cement to three (3) parts of sand.
- (f) All surfaces shall be well brushed clean and wetted before applying each coat of plaster or render. The first coat of plaster or render shall be kept damp continually before applying the second coat.
- (g) Plastering shall consist of two coats; with total thickness 20 mm thick. The undercoat shall be 12 mm thick, roughened adequately to form a key for the final coat. The finished coat shall be trowelled with proper metal floats to an even and smooth surface. All arises, internal angles, external angles, etc. shall be straight, plumb, slightly rounded to 3 mm radius and with neatly formed mitres.

G8.2.2 Render

- (a) Cement mortar rendering shall consist of two coats; the undercoat shall be 12 mm thick and scored to form a key for the final coat which shall not be less than 8 mm thick. The finishing coat shall be worked with a proper wooden float, and the face left with an even and smooth surface. The finishing coat shall be applied onto the undercoat after a minimum lapse of 24 hours and in no case shall the lapse exceed three (3) days.
- (b) Cement rendering to exterior surfaces shall incorporate a waterproofing admixture. The admixture shall be a solution of colloidal resins containing several other waterproofing chemicals, both organic and inorganic and modified through a heating process.
- (c) When integrally mixed into the cement mortar, it should form an impermeable structure and provide dense waterproofing and self-cured cement rendering, reduced shrinkage cracks and resistance to sulphate and other corrosive elements
- (d) Plastering or Rendering across Dissimilar Materials
 - (i) The Contractor shall provide a galvanised mesh strip to overlap both dissimilar surfaces at their joints before interior or exterior plastering wherever brickwork butts against concrete columns, walls or beams or steel beams. Strip mesh is to be used to cover service openings in walls before plastering is carried out.
 - (ii) Stripmesh is to be of minimum 150 mm width and composed of high quality, low carbon steel.
- (e) Curing
 - (i) The finished rendering shall be cured for at least four (4) times a day and kept damp for a period of at least fourteen (14) days after completion.
- (f) Prior to commencing any plastering, sample panels 500 mm by 500 mm of each type of quality, texture and finish shall be prepared until they are approved by the S.O.. The appropriate test panel shall be retained for reference and shall represent the standard to which all plastering of the specific type of quality, texture and finish shall be made.
- (g) Defective Work and Making Good
 - (i) All cracks, blisters and other defects shall be made good by hacking out the plaster to a rectangular shape. Edges shall be undercut to form dovetailed keys. Surfaces shall be cleaned; cement slurry shall be applied and plaster reapplied to match the level of the surrounding plaster.

G8.2.3 Painting and Decorating

- (a) All exposed surfaces and internal plastered walls, unless otherwise stated shall be painted. The Contractor shall supply all labour, painting equipment, scaffolding, protective coverings and materials and shall perform all operations in connection with the painting works.

- (b) Prior to any ordering of materials, the Contractor shall submit colour charts and colour samples of all paints (primer, sealant, finishing coats), varnishes, etc. to the S.O. for approval.
 - (i) Contractor shall execute site samples and mock-up of minimum 1 sqm of the various decorative finishes or as directed by the S.O..
 - (ii) Samples for testing including destructive test shall be supplied without charge. The cost of testing shall be borne by the Contractor.
 - (iii) All samples shall have accompanying relevant test reports if required by the S.O..
 - (iv) Colours which do not match approved samples shall be rejected.
- (c) The Contractor shall provide a five-year indemnity in the joint name of the Contractor and approved Specialist based on the attached format, in favour of the Board and the assigned, against faulty materials, workmanship and premature breakdown and fading of colours. The five-year indemnity shall commence from the S.O.'s Certified Completion Date for the project.
- (d) Deliver paint materials in sealed original labelled containers, bearing the Manufacturer's name, type of paint, brand name, colour designation, and instructions for mixing and/or reducing, pot life/expiration date, and batch/ lot numbers
- (e) No paint shall be applied to surfaces structurally or superficially damp and all surfaces shall be free from condensation, dust and any extraneous matter before application. Contractor shall follow strictly to manufacturer's specification.
- (f) The Contractor shall make good at his own expense any failure or damage to appearance due to the structure of rendered or plastered surfaces rough or damp or otherwise unsuitable or non-compliance to requirements by manufacturer or requirements approved by S.O..
- (g) Testing for Moisture Content
 - (i) Test all concrete, masonry, plaster and drywall surfaces for moisture content using a reliable electronic moisture meter (protimeter). Also test latex block fillers for moisture content before application of top coats of paint. Do not apply any paint if exceeds manufacturer's recommendation.
- (h) Preparation of Surfaces for Paint
 - (i) All surfaces shall be properly prepared to S.O.'s approval.
 - (ii) Painting Plastered Surfaces
 - 1) Surfaces for painting shall be clean and dry. Any efflorescence shall be removed by first wiping with dry coarse cloth and then with a damp cloth. The surfaces shall then be left for 48 hours to ascertain if further efflorescence occurs and no paint shall be applied until efflorescence has ceased and when surfaces are thoroughly dry;

- 2) Fill all scratches, cracks, holes, depressions and voids and fill with non-shrinking grout, speckle, patching plaster or other selected patching material. Allow to dry, refill if necessary, sand smooth to provide a flush, smooth surface; and
 - 3) Plastered surfaces shall be allowed to cure for not less than 28 days and thoroughly dry before painting commence.
- (iii) Concrete surfaces are to be brushed, down to remove dirt, efflorescence and encrusted mortar, etc.
 - (iv) Remove form oil or mould release oil from concrete surfaces using a suitable solvent and then neutralize; remove all foreign matter from concrete surfaces to be painted including all grit, dirt, loose materials, mortar drippings and the like; remove all nibs, projections or protuberances from concrete surfaces to be painted.
 - (v) Patch general concrete surfaces to be painted with Portland cement grout, filling all cracks, depressions, air holes and similar voids in order to provide smooth surfaces for painting. Do not patch architectural concrete surfaces.
 - (vi) Concrete to be allowed to cure for a minimum of 60 days or till thoroughly dry prior to application of paint or coating.
 - (vii) Brick/block surfaces are to be brushed down to remove dirt, efflorescence and encrusted mortar, etc. and defective pointing cut out and made good.
 - (viii) Thoroughly clean off all grit, grease, dirt, mortar drippings and splatters. Remove nibs or projections from masonry surfaces. Fill cracks, holes and voids, with Portland cement grout or speckle, and bag wipe surface so that it has the same texture as the adjacent masonry surface.
 - (ix) Painting to brick/block surfaces shall be 7 days after erection or thoroughly dry.
 - (x) Lead, aluminium and copper surfaces shall be washed with soap and water, roughened with abrasive paper, washed with white spirit and cleaned off.
 - (xi) Zinc or galvanised surfaces shall be degreased with white spirit and washed with clean water to remove any salts.
 - (xii) Iron and steel surfaces shall have rust, millscale, welding slag and flux residue removed by wire-brushing, scraping, hammering, flame cleaning, etc.
 - (xiii) Pre-primed iron and steel surfaces shall have defective primer, rust and loose scale removed back to bare metal and be patch-primed to match existing. Remove dirt and grease from satisfactorily primed surfaces and rub down lightly.
 - (xiv) Fill drain holes dents, cracks, hollow places, open joints and other irregularities and openings in metal work to be painted with white filler and plugs suitable for the purpose. After setting, sand to a smooth, hard finish, flush with adjoining surface.
 - (xv) Red lead primer conforming to BS 2523 and BS 7956 shall be done within one hour of preparation.

- (xvi) All structural steel members are to be shot or grit blast to BS 7079. The surface shall be free from dirt and grease.
- (xvii) All structural steel members are to be sand-blasted and general surfaces are to be thoroughly prepared to receive one coat of zinc rich epoxy primer subsequently painted with one undercoat and two finishing coats of approved enamel paint.
- (xviii) All galvanized surfaces shall be cleaned and treated with one coat of mordant solution before priming. All oil, dirt, rust, etc. shall be removed before painting commences.
- (xix) Sand to remove all roughness, loose edges, splinters and brush to remove dust. Wash off grease and dirt. Fill all cracks, splits, nail holes, screw holes and surface defects with wood putty or speckling compound after the priming coat has been applied.
- (xx) Coarse grain is also to be filled and the whole rubbed down in a smooth surface with wire wool.
- (xxi) Cut out large, loose or resinous knots and plug with sound wood. Punch all nail heads well below the surface. Stop nail heads and other imperfections. Rub down timber surfaces with fine glass paper in the direction of the grain and dust off.
- (xxii) A full coat of aluminium wood primer shall be vigorously brushed in, particular care being taken to fill end grain.
- (xxiii) Fill plywood surfaces to be painted with a plaster or cellulose-based filler after priming and rub down.
- (xxiv) For surfaces to receive transparent finish, fill nail holes, cracks and defects with wood filler matching finish colour.
- (xxv) Fire-retardant treated wood or preservation treated wood, required to be painted or stained, shall be primed prior to installation with shellac or other selected sealer, as recommended by the Manufacturer.
- (xxvi) For durability of exposed timber used externally such as door/window frames and doors, red primer shall be used and shall be double-primed and then painted with one undercoat and one/two finishing coat of synthetic enamel paint.
- (xxvii) All mortar or plaster splashes, etc. shall be removed from the plasterboard by careful scraping, any oil or grease spots shall be removed with white spirit.
- (xxviii) Fill minor imperfections with the same filler used for jointing the plasterboard. Repair any damage to the paper surface in accordance with the plasterboard manufacturer's recommendations.
- (xxix) The whole surface shall be brushed down to remove dust and other loose materials. Plasterboard surfaces shall be primed with an approved alkali-resisting primer sealer.
- (xxx) Remove dust, clean and apply 1 coat of primer, prior to painting on all insulated pipe equipment and ducts outside of mechanical equipment rooms, which will remain exposed in the finished work.

- (xxxi) Surfaces for painting shall be clean and dry. Any efflorescence shall be removed by first wiping with dry coarse cloth and then with a damp cloth. The surfaces shall then be left for 48 hours to ascertain if further efflorescence occurs and no paint shall be applied until efflorescence has ceased and when surfaces are thoroughly dry.
- (xxxii) Fill all scratches, cracks, holes, depressions and voids and fill with non-shrinking grout, speckle, patching plaster or other selected patching material. Allow to dry, refill if necessary, sand smooth to provide a flush, smooth surface.
- (xxxiii) Plastered surfaces shall be allowed to cure for not less than 28 days and thoroughly dry before painting commence.

G8.2.4 Painting

- (a) Internal Wall & Ceiling Finish products to have the minimum Technical Data as below:

3	Sealer	Top Coats (2 coats)
Finish	Matt	Low sheen
Overcoat Interval	2-4 hours	2-4 hours
Theoretical Coverage (normal conditions)	14 m ² /L	13.6 m ² /L
Volume Solids	45%	41% white
<u>Composition (nominal)</u> Pigment: Binder: Solvent:	Lightfast Non-lead Pigments Styrene Acrylic Copolymer Emulsion Water	Lightfast Non-lead Pigments Acrylic Emulsion Water
Application	Brush, roller	Brush, roller
Thinning	Ready for use	Add up to 1 part of clean water to 10 parts of paint (10%)
Colour range	White	Standard Colour Card

- (i) The systems shall meet the following environmental requirements:
 - 1) VOC Content : Less than 50g/L
 - 2) N-Methyl Pyrrolidone : Not detected
 - 3) Mercury, lead, cadmium, chromium : Not detected
 - 4) Halogenated solvent, aromatic solvent
Epichlorohydrin : Not detected
 - 5) Flash point @ 61°C : No flash
- (ii) Product to have the following Characteristics of Final Coat
 - 1) Superior dirt resistant;
 - 2) Washability;
 - 3) Green Label Product certified by Singapore Environment Council; and
 - 4) Resistant to bacteria and mildew.
- (b) Polysiloxane Paint (Selected External Metal Work)
 - (i) The properties of the coatings are as follows:
 - 1) Two-pack Epoxy Zinc Rich Primer – Interzinc 52 - Coating A
 - a) A two-component, metallic zinc rich epoxy primer which complies with the composition and performance requirements of SSPC Paint 20.
 - b) Volume solids : 59%.
 - c) Salt Spray Test : ISO 9227 “Corrosion tests in artificial atmospheres -- Salt spray tests @ 35°C” - No film defects, and no rust creep at the scribe following 1000 hours exposure.
 - d) Abrasion Resistance Test : ASTM D4060 “Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser” - Average of 385 mg weight loss per 1000 cycles using CS10 wheels and a 1 kg loading.
 - e) Adhesion Test : ISO 4624 “Paints and varnishes - Pull-off test for adhesion” using portable adhesion testers - Not less than 10MPa (1450psi) when using a PAT Model GM01 hydraulic adhesion tester on 5mm thick steel.
 - f) Impact Resistance Test : ASTM D2794 “Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)” - Direct Impact Resistance – 3 Joules.

- 2) Two Pack Epoxy Micaceous Iron Oxide, Intergard 475 HS - Coating B
 - a) A low VOC, high solids, high build, two component epoxy coating. Available with conventional pigmentation, or alternatively can be pigmented with micaceous iron oxide to provide enhanced over-coating properties.
 - b) Volume Solids : 80%.
 - c) Salt Spray Test : ISO 9227 "Corrosion tests in artificial atmospheres -- Salt spray tests @ 35°C" - No film defects, and no rust creep at the scribe following 4000 hours exposure.
 - d) Abrasion Resistance Test : ASTM D4060 "Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser" – Average of 177 mg weight loss per 1000 cycles using CS10 wheels and a 1 kg loading.
 - e) Adhesion Test : ISO 4624 "Paints and varnishes -- Pull-off test for adhesion" using portable adhesion testers – Not less than 10MPa (1450 psi) when using a PAT Model GM01 hydraulic adhesion tester on 5mm thick steel.
 - f) Impact Resistance Test : ASTM D2794 "Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)" – Direct Impact Resistance 3.2 Joules.
- 3) Two-Pack Acrylic Polysiloxane Finish, Interfine 878 - Coating C
 - a) A two component, high performance, high solid inorganic hybrid, silicon based, isocyanates free acrylic polysiloxane. It shall afford extended lifetime to first maintenance when utilized as part of a high performance anti-corrosive system, offering superior gloss and colour retention when compared to typical acrylic polyurethane and catalysed acrylic topcoats. Volume solid shall be 72;
 - b) Gloss loss of the coating shall not be more than 50% when measured at an angle of 60 degrees in accordance with the measurement of specular gloss of non-metallic paint film BS3900 (part D5) 1980 (1992) and in the notes for guidance on the conditions of natural weathering test BS3900 (part F6) 1976 (1991) by Elcometer 402 ex Elcometer instruments with specific procedure of measurements; and

- c) Color level of the coating shall not change more than Delta E (DE) level; DE 3 for colours white, grey and blacks; DE 4 for blue, yellow and greens, and DE 7 for red and oranges when measured with the CMC equation using X-Rite SP64 Portable Spectrophotometer ex Elcometer Instruments with specific procedures of measurements.
- 4) Two-pack Surface Tolerant Epoxy Primer, Interseal 670 HS - Coating D
 - a) A low VOC, two-component high build, high solids surface tolerant epoxy maintenance coating. Available in an aluminium pigmented version to provide additional anti-corrosive barrier protection;
 - b) Volume Solids : 82%.
 - c) Cathodic Disbondment Test : ASTM G8 "Cathodic Disbonding of Pipeline Coatings" Method A – Typically less than 3mm disbondment following 30 days exposure.
 - d) Immersion Test : ISO/DIS 2812 Part 2 "Paints and varnishes – Determination of resistance to liquids – Part 2: Water immersion method @ 40°C" – No film defects after 1-year exposure.
 - e) Salt Spray Test : ISO 9227 "Corrosion tests in artificial atmospheres -- Salt spray tests @ 35°C" – No film defects, and an average of 1 mm rust creep at the scribe after 3000 hours exposure.
 - f) Abrasion Resistance Test : ASTM D4060 "Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser" – Average of 259 mg weight loss per 1000 cycles using CS17 wheels and 1 kg loading.
 - g) Adhesion Test : ISO 4624 "Paints and varnishes - Pull-off test for adhesion" using portable adhesion testers – Not less than 7MPa (1015 psi) when using a PAT Model GM01 hydraulic adhesion tester on 5mm thick steel.

- h) Impact Resistance Test : ISO 6272 “Paints and varnishes – Rapid-deformation (impact resistance) tests – Falling-weight test” – Pass at <25 cm height with a 2 kg load.
- (c) Performance Requirements
 - (i) The design life of the coating system shall have a durability of 10 years against corrosion and a durability of 5 years against colour and gloss retention before first major maintenance for exposed structure with specified regular minor maintenance program.
 - (ii) All steel surface shall comply with the proposal for the protective system provided.
 - (iii) Assessment of the level of coating failure before the first major maintenance in accordance with EN ISO 12944 standard.
- (d) Painting to Plasterboard and Other Board Surfaces
 - (i) Plasterboard surfaces shall be primed with an approved alkali resisting primer sealer and two coats of approved emulsion paint.
- (e) Painted Traffic Signs and Road Lines
 - (i) Unless otherwise specified, all paints used for painted traffic signs and road lines on roads, kerbs, etc. shall be durable, capable of subjecting to heavy and continuous traffic. Only thermoplastic paint complying with the general specifications shall be used and subject to the S.O.’s approval.
- (f) Execution
 - (i) Examination
 - 1) The Contractor shall examine areas and conditions under which painting work is to be applied and correct conditions detrimental to proper and timely completion of work. Do not proceed with work until unsatisfactory conditions have been corrected;
 - 2) Starting Work constitutes acceptance of the existing conditions and the Contractor shall, at his own expense, be responsible for correcting all unsatisfactory and defective Work encountered;
 - 3) Measure moisture content of surfaces using an electronic moisture meter. Do not begin application of coatings unless moisture content of surfaces is below the following maximum values or any more stringent values specified by the coating system manufacturer:
 - a) Gypsum wallboard: 12 percent;
 - b) Plaster: 12 percent;
 - c) Masonry surfaces: 12 percent;
 - d) Wood surfaces: 15 percent;
 - e) Vertical concrete surfaces: 12 percent; and
 - f) Horizontal concrete surfaces: 8 percent.

- 4) Verify that site environmental conditions are appropriate and substrates are in proper condition to receive work of this section; and
 - 5) Verify that shop applied primers are compatible with specified finish coats.
- (ii) Preparation
- 1) Unless specifically stated to the contrary, the descriptions of Painting Work shall be understood to include all preparatory work, and priming, required and necessary to produce a first class finish, free from all blemishes, brush marks, blisters and weeping and the following definitions are to apply where preparation is to be carried out:
 - a) “Wash down” shall mean either:
 - (iii) The removal with clean water of all dirt, etc. not absorbed into the surface of a material not previously decorated, or
 - (iv) The removal with clean water of existing lime wash, non-washable distemper or similar material not absorbed into the decorated surfaces, or
 - (v) The removal of dirt, etc. from and cleaning down of, existing washable distempered, cement painted, synthetic painted and similarly decorated surfaces with sugar soap powder mixed with water, followed by further applications of clean water.
 - 1) “Broom down” shall mean the thorough dry brushing of any surface with a stiff broom or brush so as to remove all cobwebs, dust or loose materials.
 - 2) “Scrape” shall mean the removal of all existing coats of paint, lime wash, colourwash, distemper, etc. by scraping tools without the use of chemical solvents or heat and without damage to the underlying material. Any damage so caused shall be made good. The surface is to be washed down on completion.
 - 3) “Wire brush” shall mean the thorough brushing of the surface with a stiff wire brush.
 - 4) “Strip” shall mean the complete removal, without damage to the underlying surface, of all existing coats of lime wash, distemper, paint or other decorative material by means of washing and scraping together with the use of chemical solvents or heat, if necessary. After the use of chemical solvents the surfaces is to be washed down and neutralised in accordance with the Manufacturer’s recommendation.
 - 5) “Stop” to plaster shall mean carefully trimming the edges of all holes, cracks or crevices of any description and filling with an approved proprietary brand of filler or with Plaster of Paris to produce an even, flat surface, and touching up all patches with a coat of sealer prior to repainting. To apply 10mm PVC stopping beads to door frame as shown in door details
 - 6) “Stop” to woodwork shall mean the cleaning out and filling of all holes, cracks and crevices, etc.

- 7) "Knot" to metal pipes shall mean the application of one coat of patent knotting to surface of any bitumen coated pipe.
 - 8) "Knot" to woodwork shall mean the application of a sufficient number of coats of "knotting" over all knots in the timber to prevent the bleeding of resin, etc. through the subsequent decoration.
 - 9) "Rub down" shall mean the rubbing of newly prepared surfaces, or surfaces of existing paint remaining after preparation for redecoration, with approved waterproof glass paper, pumice stone or similar, to give a true and keyed surface for new paint.
 - 10) "Fill" shall mean filling of grain with approved surface filler.
 - 11) Do not paint over dirt, rust, scale, grease, moisture, scuffed surfaces, and other conditions otherwise detrimental to formation of a durable paint film.
 - 12) Perform preparation and cleaning procedures in accordance with paint manufacturer's printed instructions for each particular substrate condition.
 - 13) Provide barrier coats over incompatible primers or remove and prime as required.
 - 14) Remove finish ironmongery, hardware, hardware accessories, machined surfaces, plates, lighting fixtures, and similar items in place and not to be painted or provide surface applied protection prior to surface preparation and painting operations. Reinstall all removed items after completion of paint work.
 - 15) Clean surfaces to be painted before applying paint of surface treatment. Remove oil and grease prior to mechanical cleaning.
- (vi) Correct defects and clean surfaces affecting work of this section. Remove existing coatings that are flaking or otherwise in unacceptable condition to receive paint.
- 1) Seal with shellac or other coating acceptable to paint manufacturer any marks or defects that might bleed through paint finishes.
 - 2) Remove mildew from impervious surfaces by scrubbing with solution of tri-sodium phosphate and bleach. Rinse with clean water and allow substrate to dry.
 - 3) Ferrous Metals: Clean ferrous surfaces that are not galvanized or shop-coated, free of oil, grease, dirt, loose mill scale and other foreign substances by solvent or mechanical cleaning. Surface cleaning to SA2.5 and followed by solvent cleaning to SSPC-SP1 as a minimum standard.
 - 4) Touch-up shop-applied prime coats, where damaged or bare. Clean and touch-up with same type of shop primer.
 - 5) Galvanized Surfaces: Clean free of oil and surface contaminants with non-petroleum based solvent. Apply coat of etching primer if required by Paint Manufacturer.

- 6) Plaster and Drywall: Fill hairline cracks, small holes, and imperfections with patching compound, then smooth patches to match adjacent surfaces.
 - a) Determine alkalinity and moisture content of surfaces to be painted by performing appropriate tests.
 - b) If surfaces are found to be sufficiently alkaline to cause blistering and burning of finish paint, correct condition before application of paint.
 - c) Do not paint over surfaces where moisture content exceeds that permitted in Manufacturer's printed instructions.
- 7) Cementitious Materials: Prepare cementitious surfaces of concrete, brick, and cement plaster to be painted by removing efflorescence, chalk, dust, dirt, grease, oils, and by roughening as required to remove glaze.
 - a) Determine alkalinity and moisture content of surfaces to be painted by performing appropriate tests.
 - b) If surfaces are found to be sufficiently alkaline to cause blistering and burning of finish paint, correct condition before application of paint.
 - c) Do not paint over surfaces where moisture content exceeds that permitted in Manufacturer's printed instructions.
 - d) Clean floor surfaces scheduled to be painted with a commercial solution of muriatic acid, or others etching cleaner. Flush floor with clean water to neutralize acid and allow drying before painting.
- 8) Wood: Clean wood surfaces to be painted of dirt, oil, and other foreign substances with scrapers, mineral spirits, and sandpaper, as required. Sandpaper smooth those finished surfaces exposed to view and dust off. Scrape and clean small, dry, seasoned knots and apply a thin coat of white shellac or other recommended knot sealer, before application of priming coat. After priming, fill holes, and imperfections in finish surfaces with putty or plastic wood-filler. Sandpaper smooth when dried.
 - a) Prime, stain, or seal wood required to be job-painted immediately upon delivery to job. Prime edges, ends faces, undersides, and backsides of such wood, including cabinets, counters, cases, trims, skirting, and panelling.
 - b) Use spar varnish for back priming transparent finish surfaces.
 - c) Back prime panelling on interior partitions only where masonry, plaster, or other wet wall construction occurs on backside.
 - d) Seal tops, bottoms, and cut-outs of unprimed wood doors with a heavy coat of varnish or equivalent sealer immediately upon delivery to Project Site.

- 9) Shop-Primed Ferrous Metals: Clean all bare and abraded areas to bright, bare metal. Spot prime with specified primer at once.

(g) Application

- (i) Carry out the works strictly in accordance with each paint manufacturer's recommendations and specified method.
- (ii) Apply each coat at proper consistency. Protect other surfaces from paint and damage. Repair damage as a result of inadequate or unsuitable protection.
- (iii) Each coat of paint is to be slightly darker than preceding coat unless otherwise approved by the Architect. Sand lightly between coats as required in order to achieve the specified finish.
- (iv) Do not apply finishes on surfaces that are not dry.
- (v) The Paint Schedule provides for a minimum three-coat application, U.N.O. The intent of these Specifications is to provide 100% protective coverage for all exposed surfaces scheduled for painting.
 - 1) Additional coat or coats may be directed by Architect to give complete coverage and uniform appearance;
 - 2) Primer may be omitted for items shop primed; and
 - 3) Paint inside surfaces of sheet metal items and assemblies.
- (vi) Apply additional coats when undercoats, stains, or other conditions show through final coat of paint until paint film is of uniform finish, colour, and appearance. Give special attention to ensure that surfaces, including edges, corners, crevices, welds, and exposed fasteners, receive a dry film thickness equivalent to that of flat surfaces.
- (vii) Minimum Coating Thickness: Apply materials at not less than the Manufacturer's recommended spreading rate. Providing a total dry film thickness (DFT) of the entire system as recommended by the Manufacturer, or as specified herein for stricter DFT.
- (viii) Prime Coats: Before application of finish coats, apply a prime coat of material as recommended by the Manufacturer to material that is required to be painted or finished and has not been previously prime coated. Recoat primed and sealed surfaces where evidence of suction spots or unsealed areas in first coat appears, to assure a finish coat with no burn through or other defects due to insufficient sealing.
- (ix) Pigmented (Opaque) Finishes: Completely cover to provide an opaque, smooth surface of uniform finish, colour, appearance, and coverage. Cloudiness, spotting, laps, brush marks, runs, sots, or other surface imperfections are not acceptable.
- (x) Transparent (Clear) Finishes: Use multiple coats to produce a glass-smooth surface film of even lustre. Provide a finish free of laps, cloudiness, colour irregularity, brush marks, runs, sags, orange peel, nail holes or other surface imperfections.

- (xi) Completed Work: Match approved samples for colour, texture, and coverage. Remove, refinish, or repaint all work not in compliance with specified requirements until the completed work meets the approval of the Architect.
- (xii) Provide adequate fresh air, exhausting and other measures to ensure that specified indoor air quality during and after Work of this Section is met. All Volatile Organic Compounds, air pollutants and other toxins released during Work of this Section shall be purged in strict accordance with the requirements of the Contract Documents.
- (xiii) Internal Wall & Ceiling Finish
 - 1) Sealer:
 - a) Apply by roller/ brush 1 coat of Sealer a styrene acrylic water-based sealer to conforms to Singapore Green Labelling Scheme (032 038).
 - b) No Thinning, Ready for use.
 - c) Overcoat Interval : 2-4 hours
 - d) Coverage (normal conditions) : 14 m²/L
 - 2) Top Coats:
 - a) Apply by roller/ brush 2 coats of 100% acrylic paint interior walls and ceilings with Anti-Bacteria. Soft sheen finish, low Volatile Organic Compound (VOC) emission and low odour, stain resistance and scrubbability, with good opacity, to conforms to Singapore Green Labelling Scheme (032 038).
 - b) Overcoat Interval : 2-4 hours
 - c) Coverage (normal conditions) : 13.6 m²/L
 - d) Thinning up to maximum 1 part of clean water to 10 parts of paint (10%).
- (xiv) Polysiloxane Paint for External Steel Work (Selected)
 - 1) Carbon Steel
 - a) Shop Applied

All surface shall be free from oil, grease and salt contamination before blasting. Blast clean to SA2.5 (ISO 8501-1) or SSPC-SP10, surface amplitude 50 to 75 micron. If oxidation has occurred between blasting and application, the surface should be reblasted to the specified visual standard.

- | | | |
|----|---|----------------------------|
| b) | Minimum Dry Film | Thickness (Microns) |
| | Two-pack epoxy zinc rich primer
Interzinc 52 (Coating A) | 75 |
| | Two-pack epoxy micaceous oxide
Intergard 475 HS (Coating B) | 250 |
| | Two-pack acrylic polysiloxane finish
Interfine 878 (Coating C) | 75 |
| c) | Site Applied/Repair Procedure | |
| | Power tool clean to Steel Structures Painting Council's Specification SSPC-SP3-63 "power tool cleaning" to give surface finish St3 grade to Swedish Standard. | |
| d) | Minimum Dry Film | Thickness (Microns) |
| | Touch up with two-pack surface tolerant epoxy primer interseal 670 HS (Coating D) | 100 |
| | Touch up with two-pack epoxy micaceous oxide Intergard 475 HS (Coating B) | 125 |
| | Touch up with two-pack acrylic polysiloxane finish Interfine 878 (Coating C) | 75 |
- (xv) Mechanical, Electrical Equipment and Installations
- 1) Replace identification markings on mechanical, electrical and other equipment when painted over or spattered.
 - 2) Paint gas piping prior to installation.
 - 3) Paint exposed pipes, conduit and electrical equipment occurring in finished areas where it will be exposed to the public. Colour and texture to match adjacent surfaces.
 - 4) Paint both sides and edges of plywood backboards for electrical equipment and other applications before installing backboards and mounting equipment on them.
 - 5) Refer to Contract Documents for other painting requirements affecting mechanical and electrical equipment and installations.
- (xvi) Clean up prior to completion
- 1) As Work proceeds, and upon completion, promptly remove paint where spilled, splashed, or spattered. During progress of Work keep premises free from any unnecessary accumulation of tools, equipment, surplus materials, and debris.
 - 2) Protect work adjacent to painting operations from paint spatters and spills. Immediately remove paint that falls on finished surfaces not scheduled to receive paint, using materials and techniques that will not damage affected surfaces

- 3) Keep project premises free of painting-related debris. Collect material that may constitute a fire hazard, place in closed metal containers, and remove daily from site.
 - 4) Upon completion of Work leave premises neat and clean.
 - 5) Legally transport and dispose of all paints, solvents and other painting products in accordance with local Environmental Protection regulations.
 - 6) Prior to the issuance of the Completion Certificate clean or replace all air filters in air inlets and outlets after purging of the Works.
- (xvii) Protection
- 1) Protect other surfaces from paint and damage. Repair damage as a result of inadequate or unsuitable protection.
 - 2) Install "Wet Paint. Do not touch." signs in English and barrier tapes around painted areas and items.
- (xviii) Paint Finishing Schedule
- 1) To provide Paint Schedule for minimum three-coat application, unless otherwise stated. A fourth coat or further coats may be required as directed to provide complete coverage and uniform appearance. Omit primer for items shop primed, unless primed surfaces show signs of poor adhesion and cohesion, defects and other detrimental conditions. Re-prepare and prime again, such surfaces as part of the Work.

G8.2.5 Architectural Paint Systems

*Coverage is listed as total MDFT (minimum dry film thickness) in mm and Per-coat determinations are listed as MDFTPC.

- (a) Steel Exterior Acrylic - Semi-Gloss
 - (i) Surface Preparation: hand tool cleaning, SSPC SP-2.
 - (ii) Primer: universal alkyd primer, one coat, 0.076 MDFT or direct-to-metal acrylic primer, 0.064 MDFT.
 - (iii) Finish Coats: acrylic latex, semi-gloss, 0.033 MDFTPC.
- (b) Steel Interior Acrylic - Semi-Gloss
 - (i) Surface Preparation: hand tool cleaning, SSPC SP-2.
 - (ii) Primer: direct-to-metal acrylic primer, one coat, 0.064 MDFT.
 - (iii) Finish Coats: acrylic latex, semi-gloss, two coats, 0.033 MDFTPC.
- (c) Steel Interior Alkyd - Semi-Gloss
 - (i) Surface Preparation: hand tool cleaning, SSPC SP-2.
 - (ii) Primer: universal alkyd primer, one coat, 0.076 MDFT.
 - (iii) Finish Coats: alkyd enamel, semi-gloss, two coats, 0.058 MDFTPC.

- (d) Steel Structure/Galvanized Metal Deck, Dryfall - Alkyd
 - (i) Surface Preparation: solvent cleaning, SSPC SP-1.
 - (ii) Primer: direct-to-metal wash primer, one coat, 0.025 MDFT.
 - (iii) Finish Coats: alkyd dryfall, eggshell, one to two coats, 0.051 MDFTPC.
- (e) Steel Structure/Galvanized Metal Deck, Dryfall - Latex
 - (i) Surface Preparation: solvent cleaning, SSPC SP-1.
 - (ii) Primer/Finish Coats: water-based acrylic dryfall, eggshell, two coats, 0.102 MDFTPC.
- (f) Galvanized Steel - Interior - Acrylic
 - (i) Surface Preparation: solvent cleaning, SSPS SP-1 or brushoff-blast cleaning, SP-7 if required to promote adhesion.
 - (ii) Primer: direct-to-metal acrylic primer, one coat, 0.064 MDFT.
 - (iii) Finish Coats: acrylic latex, semi-gloss, two coats, 0.041 MDFTPC.
- (g) Concrete Exterior/Interior Acrylic - Flat
 - (i) Surface Preparation: concrete.
 - (ii) Primer: acrylic masonry primer, one coat 0.091 MDFT.
 - (iii) Finish Coats: acrylic latex flat, two coats, 0.038 MDFTPC.
- (h) Concrete Exterior/Interior Acrylic - Semi-Gloss
 - (i) Surface Preparation: concrete.
 - (ii) Primer: acrylic masonry primer, one coat 0.091 MDFT.
 - (iii) Finish Coats: acrylic latex semi-gloss, two coats, 0.033 MDFTPC.
- (i) Concrete Unit Masonry Exterior/Interior Acrylic - Flat
 - (i) Surface Preparation: masonry.
 - (ii) Primer: block filler, one coat, 0.203 MDFT.
 - (iii) Finish Coats: acrylic masonry coating, two coats, 0.033 MDFTPC.
- (j) Concrete Unit Masonry Exterior/Interior Acrylic - Semi-Gloss
 - (i) Surface Preparation: masonry.
 - (ii) Primer: block filler, one coat, 0.203 MDFT.
 - (iii) Finish Coat: acrylic masonry coating, two coats, 0.033 MDFTPC.
- (k) Concrete or Concrete Unit Masonry Exterior/Interior - Opaque Stain
 - (i) Surface Preparation: concrete or masonry as applicable.
 - (ii) Primer: none.
 - (iii) Finish Coats: acrylic stain, two coats to achieve full penetration and coverage.

- (l) Concrete or Concrete Unit Masonry Exterior - Elastomeric Coating
 - (i) Surface Preparation: concrete or masonry as applicable.
 - (ii) Primer: masonry block filler, if necessary to fill surface voids.
 - (iii) Finish Coats: single-component elastomeric acrylic coating, two coats, 0.127 MDFTPC.
- (m) Gypsum Board Interior Acrylic - Flat
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: latex primer, one coat, 0.041 MDFT.
 - (iii) Finish Coats: acrylic latex, flat, two coats, 0.036 MDFTPC.
- (n) Gypsum Board Interior Acrylic Latex - Eggshell
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: latex primer, one coat, 0.041 MDFT.
 - (iii) Finish Coat: acrylic latex, eggshell, two coats, 0.041 MDFTPC.
- (o) Gypsum Board Interior Acrylic Latex - Semi-Gloss
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: latex primer, one coat, 0.041 MDFT.
 - (iii) Finish Coats: acrylic latex, semi-gloss, two coats, 0.033 MDFTPC.
- (p) Gypsum Board Interior Acrylic Latex - Gloss
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: latex primer, one coat, 0.041 MDFT.
 - (iii) Finish Coats: acrylic latex, gloss, two coats, 0.033 MDFTPC.
- (q) Gypsum Board Water-Based Epoxy
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: latex primer, one coat, 0.036 MDFT.
 - (iii) Finish Coats: water-based catalyzed epoxy, two coats, 0.076 MDFTPC.
- (r) Gypsum Board Interior - Textured Coating
 - (i) Surface Preparation: gypsum board.
 - (ii) Primer: gypsum board primer/sealer, one coat per manufacturer's recommended coverage.
 - (iii) Intermediate Coat: polychromatic, terpolymer coating, one coat per manufacturer's recommended coverage.
 - (iv) Finish Coat: optional, clear epoxy semi-gloss coating.

- (s) Wood Stained - Exterior
 - (i) Surface Preparation: wood.
 - (ii) Primer: none.
 - (iii) Finish Coats: acrylic polyurethane solid-color stain, two coats for full coverage and penetration.
- (t) Wood Stained and Varnished - Exterior/Interior
 - (i) Surface Preparation: wood.
 - (ii) Stain: oil-based transparent stain.
 - (iii) Intermediate Coat: sanding sealer, one coat, 0.025 MDFT.
 - (iv) Finish Coat: oil varnish, satin, two coats, 0.043 MDFTPC.
- (u) Wood Transparent Finish - Interior
 - (i) Surface Preparation: wood.
 - (ii) Primer: none.
 - (iii) Finish Coats: water-based polyurethane, satin three coats, 0.033 MDFTPC.
- (v) Wood Opaque Finish Exterior Acrylic - Semi-Gloss
 - (i) Surface Preparation: wood.
 - (ii) Primer: exterior alkyd wood primer, one coat, 0.058 MDFT.
 - (iii) Finish Coats: exterior acrylic latex, semi-gloss, two coats, 0.033 MDFTPC.
- (w) Wood Opaque Finish, Semi-Gloss, Interior - Alkyd
 - (i) Surface Preparation: wood.
 - (ii) Primer: alkyd wood primer, one coat, 0.041 MDFT.
 - (iii) Finish Coats: alkyd enamel, semi-gloss, two coats, 0.033 MDFTPC.
- (x) Wood Opaque Finish, Semi-Gloss, Interior - Acrylic
 - (i) Surface Preparation: wood.
 - (ii) Primer: latex wood primer, one coat, 0.041 MDFT.
 - (iii) Finish Coats: latex, semi-gloss, two coats, 0.033 MDFTPC.
- (y) Ferrous Metal, Gloss, Interior - High Solids Industrial Enamel
 - (i) Surface Preparation: metal, commercial blast, SP-6.
 - (ii) Primer: universal alkyd metal primer, one coat, 0.102 MDFT.
 - (iii) Finish Coat(s): high solids, low VOC, industrial alkyd enamel, gloss sheen; two coats, 0.102 MDFTPC.
- (z) Ferrous Metal, Semi-Gloss, Interior - Water Based Epoxy Acrylic
 - (i) Surface Preparation: metal, commercial blast, SP-6.
 - (ii) Primer: universal alkyd rust inhibitive metal primer; one coat, 0.076 MDFT.
 - (iii) Finish Coats: water-based epoxy acrylic; two coats, 0.076 MDFTPC.

- (aa) Concrete Walls, Ceilings, and Columns; Semi-Gloss, Interior - Water Based Epoxy Acrylic
 - (i) Surface Preparation: concrete.
 - (ii) Fill Coat: water based epoxy block filler; as required to fill voids to provide smooth surface.
 - (iii) Primer/Finish: water-based epoxy acrylic, semigloss; two coats, 0.051 to 0.102 MDFTPC.
- (bb) Concrete Floor; Semi-Gloss, Interior - Polyamide epoxy
 - (i) Surface Preparation: concrete.
 - (ii) Primer: high solids, moisture tolerant polyamide epoxy, one coat, 0.127 to 0.203 MDFT.
 - (iii) Finish Coats: high solids, semi-gloss, polyamide epoxy, one coat, 0.127 to 0.203 MDFT.

G8.2.6 Delivery, Storage and Handling

- (a) All materials for painting shall be delivered to the Site in the original sealed containers supplied by the manufactures and shall be carefully stored to minimise exposure to high temperatures in accordance with manufacturer's recommendation. Storage for paint shall carry warning signs both inside and outside in all the official local languages reading "Strictly No Smoking" and "Flammable".
- (b) Maintain neat, clean conditions in storage area.
- (c) Store and mix materials in defined site locations. Do not open manufacturer's sealed containers until materials and colours have been accepted.
- (d) Close containers and remove flammable materials from the premises. Leave no materials open.
- (e) Paints other than water-based and bituminous paints are to be delivered in containers of not more than 5 litres capacity.

G8.2.7 Protection

- (a) All surfaces of the painted work, either priming or finishing paint, shall be protected throughout the duration of the Contract.
- (b) Before commencing work, adjoining areas which are not to be painted shall be adequately protected.
- (c) Display "wet paint" signs and provide adequate protective barriers whenever and wherever necessary.
- (d) Mask all adjoining surfaces when spray painting is being applied.
- (e) Any material which is marked, stained or mechanically damaged shall be cleaned off and made good to the approval of the S.O. or if in his opinion, cleaning and making good is impractical, will be completely redecorated.

G8.2.8 Tiling - General

- (a) Submissions
 - (i) Submission of floor, wall and special purpose tiles for the S.O.'s approval shall be carried out immediately after the award of the Contract. Any delays by the Contractor shall be entirely at the Contractor's responsibility.
 - (ii) The S.O. may select the tiles from the Contractor's submission or failing which, may subsequently select the tiles from other suppliers.
- (b) Delivery
 - (i) All tiles shall be delivered to the site in their original unopened containers with all labels intact and legible at time of use.
- (c) Samples and Mock-Ups
 - (i) Samples of tiles shall be submitted free of cost to the S.O. for approval.
 - (ii) The contractor shall loose lay a mock-up (for all types of tiles) of approximately 0.5 m², for S.O.'s approval prior to laying. The exact location, colours/pattern of tiles shall be directed by the S.O.. The mock-up shall be executed to the complete satisfaction of the S.O.. Upon acceptance of the mock-up by the S.O., the Contractor shall be required to submit full tiling layout drawings to a minimum of 1:100 scale.
 - (iii) The Contractor shall submit 1 set of samples of the approved tiles mounted on display boards to the S.O. at the time of or immediately after the approval of the loose lay mock-up.
 - (iv) All tiles delivered to the site will be gauged against the approved samples for evenness of colour and quality.
- (d) Test Reports
 - (i) The Contractor shall submit local test report from approved testing laboratories to show the proposed tiles comply in all respect with the specification.
- (e) Tests
 - (i) Notwithstanding the submission of test reports, the Contractor shall also allow for the cost of providing samples, transport, air freight (if any), testing fees and all other charges and expenses incurred in sending 3 lots each of the tiles from each consignment to ENTERPRISE SINGAPORE or other approved laboratories separately for testing in 20 pieces per lot selected at random at the discretion of the S.O..
 - (ii) The Contractor shall provide the services of a fork-lift and any other necessary equipment for the sampling of tiles. Samples will be picked at random by the S.O. or his authorised representative(s) and the Contractor shall ensure that any pallet or pallets be opened for this purpose.
 - (iii) Tiles selected at random will be despatched separately in 3 lots of 20 pieces each to Enterprise Singapore or other approved laboratories for testing at Contractor's own expense.

- (iv) Tiles shall be tested in accordance with methods and procedures adopted by ENTERPRISE SINGAPORE or other approved bodies.
 - (v) If the tiles tested fail to meet the Specification, the entire pallet or the entire tiled area shall be removed from the site and made good at the Contractor's own expense.
- (f) Test Report
 - (i) The Contractor shall submit local test report from PSB to show that the proposed tiles comply in all respect with the specification.
- (g) Delivery
 - (i) All tiles shall be delivered to the site in their original unopened containers with all labels intact and legible at time of use.
- (h) Packaging and Storage
 - (i) All tiles shall be packed in properly labelled wooden pallets complete with plastic covers.
 - (ii) All tiles shall be kept in batches, numbered in accordance with the number of the containers and/or pallets.
 - (iii) All tiles shall be stored under cover in such manner as to prevent damage and contamination, at designated locations on site.
- (i) Colour
 - (i) The Contractor shall, provide full range patterns, colours and textures of the tiles. The S.O. shall have absolute prerogative in the selection of tiles on their particular patterns, colour ranges and textures and there shall be no additional cost to the Board.
 - (ii) To ensure consistent tonal quality, tiles should preferably be manufactured in a single process irrespective of delivery schedule. The Contractor shall ensure that tiles of the same type in any one area or in two or more adjoining areas shall be the same and shall show no sign of change in quality, colour, hue, shade, tone, texture, etc.
- (j) Accessories
 - (i) The Contractor shall provide all accessory tiles in matching size and colour to trim round all water pipes and any other service pipes inclusive of concrete kerbs, windows and any other openings to the satisfaction of the S.O.. The Contractor is deemed to have included the cost of accessory tiles in his tender sum
- (k) Vitreous Bull Nosed Non-Slip Tiles
 - (i) Where stairs are specified as requiring a tiled finish, the Contractor shall supply and install 150 x 75 x 19 mm thick full vitreous bull-nosed non-slip tiles bedded and jointed in cement mortar to treads in accordance with the specification for the application for floor tiles.
- (l) Aluminium Dividing Strips

- (i) The Contractor is to supply and fix 25 x 2 mm thick aluminium dividing strips bedded on edge and including fair joints when the screed length exceeds 2.5 m in both directions and between different floor finishes or as instructed on site by the S.O..
- (m) Tiles for Maintenance Stock
 - (i) Supply and deliver only ceramic wall tiles all as specified and as approved by the S.O..

	Percentage for Area Applied	Maintenance	Remarks
1)	Up to 300m ²	2%	Max. 6m ²
2)	Over 300 to 1,200m ²	1%	Min. 6m ² Max. 12m ²
3)	Over 1,200 to 8,000m ²	1/4%	Min. 12m ² Max. 20m ²
4)	Over 8,000 to 20,000m ²	1/8%	Min. 20m ² Max. 25m ²
5)	Over 20,000m ²	1/10%	Min. 25m ² Max. 40m ²

- (ii) The maintenance stock shall be properly protected and upon the direction of the S.O., be passed on to the Board for keeping. Certified vouchers of the tiles by the recipient shall be forwarded to the S.O. for record.

G8.2.9 Internal Wall Tiling

- (a) General
 - (i) Internal wall tiles shall be "Italia" tiles or approved equivalent, coloured and sized 200 mm x 200 mm in plan unless specified otherwise. All wall tiles shall be of metric modular size.
- (b) Properties
 - (i) Wall tiles shall be 8 mm thick first quality fully glazed tiles to SS 57 or equivalent with the following characteristics:
 - Water Absorption in weight 5% maximum
 - Modulus of Rupture 25 N/mm² minimum
 - Moh's Hardness 5 minimum
 - Chemical resistance Macroscopic surface examination shall not reveal any visible change
 - Resistance to staining Class I (minimum)
 - Colour Fastness Tiles to be colour-fast
 - (ii) Tiles shall be clean, uniform gauge, texture and colour and free from all defects. The Contractor shall submit samples of at least 20 different patterns and colours of the glazed wall tiles to the S.O. for approval and no order shall be placed with the supplier until such approval has been given.

- (iii) Wall tiles shall be supplied and installed complete with all corner angles, matched coved skirting tiles or other purpose made accessory tiles. Purpose made internal and external angle tiles shall be used as required.
- (c) Preparation
 - (i) Wall tiles shall be laid on top of previously cement rendered surfaces in the case of masonry substrate as specified herein and on top of the bare concrete in the case of structural concrete substrate and shall be laid with a full bed of water resistant tile adhesive as recommended by the manufacturer. Tiles shall not be fixed until surfaces are thoroughly cured and completely dry.
- (d) Laying
 - (i) Tiling shall be set out in an uniform and symmetrical manner. All tiles shall be cut neatly and edges ground smooth to ensure bonding. Tiles shall be carefully cut and fitted around all obstructions such as pipes, etc. Small make-up pieces should generally be avoided and for margins, half tile or wider shall be used where practicable.
 - (ii) Joints shall be 2 mm wide, +/- 1 mm. After fixing, joints shall be finger pointed flush with white Parian cement.
 - (iii) Tile shall be cleaned upon completion and all surfaces left tidy and to the satisfaction of the S.O..

G8.2.10 Melded Fabric

- (a) General
 - (i) The Contractor shall supply and install melded fabric which is a non-woven thermally bonded blend of polyester and nylon for the areas as specified in the Schedule of Internal Building Finishes. The melded fabric shall be insect and mildew resistant and have an anti-static agent incorporated during production. Colour choice should be subject to the approval of the S.O..
- (b) Properties
 - (i) The melded fabric shall consist of the following:
 - 1) Widths 1.4 m - 1.83 (selected colours)
 - 2) Roll Length Average 50 m
 - 3) Fabric Thickness 3-4 mm
 - 4) Fabric Weight AS 2001 275 gm/m²
 - 5) Composition 60% polyester: 40% nylon
 - 6) Tensile Strength
 - a) ASTM 1682 (strip) Transverse - 290 N/50 mm
Longitudinal - 180 N/50 mm
 - 7) Tear Strength
 - a) ASTM 2263 (Trapezoid) Transverse - 220 N
Longitudinal - 130 N

- 8) Abrasion Resistance
 - a) Taber, H10 1500 cycles
moderate wheels, appearance change to 1 kg load surface only.
 - b) Stoll AS 2687 Heavy duty rating after 5,000 cycles B-84.
 - c) Martindale Rub through point to exceed AS 2687
A-84 100,000 cycles. Appearance change after 40,000 cycles.
- 9) Fastness to Light AS 2001-4-21-1970, 4/5 rating. (colour fade). Not suitable for exterior use.
- 10) Light Reflectance Values 14% - 31% inclusive (fluorescent light source).
- 11) Fire hazard properties:

Ignitability index	11	Range 0-
20 Spread of Flame index	0	Range 0-
10 Heat evolved index	1	Range 0-10
Smoke developed index	4	Range 0-10

(c) Installation

- (i) For semi-porous surfaces such as timber or plasterboard, a thick water base PVA latex adhesive is to be used. Very porous surfaces should be sized with diluted adhesive and allowed to dry for 24 hours prior to proceeding. One litre of adhesive shall bond approximately 4 square metres of fabric.
- (ii) Wall surfaces should be clean & dry with all cracks and joints fully filled and sanded prior to commencement. Adhesives should be applied to the wall and only sufficient adhesive should be applied for one fabric piece at time. Care should be taken to avoid getting adhesive onto the fabric surface. Should this occur, the adhesive must be removed before it dries.
- (iii) Each strip shall be hung in the same direction, to avoid apparent colour changes.
- (iv) Batch numbers shall be checked and butt joints between different batch numbers shall be avoided.
- (v) The fabric shall be hung from the top of the wall, gradually working down. Overlaps in the fabric shall be cut through with a blade to achieve a good join.

G8.2.11 Floor Finishes

- (a) Floor finishes shall not be laid prior to completion of plastering and wall tiling. Where pipes are brought through floors, the floor finish shall be neatly finished level around metal sleeves and shall not be finished against the pipes.
- (b) Where necessary to grade finished surface to fall, screed thickness may be adjusted with approval from S.O..

- (c) Preparation of Receiving Surfaces
 - (i) Concrete surfaces specified to be finished with cement mortar screed, granolithic screed, ceramic or homogenous tiles, or other applied finishes, shall be thoroughly roughened, brushed with wire brushes to expose a fresh surface, swept clean and washed with clean water until well soaked. A thin layer of stiff cement grout shall then be well brushed into the roughened surface a few minutes before laying the applied finish.
- (d) Where floor hardener and dust proofer are specified or indicated in the Schedule of Finishes, floor hardener may be applied by a “dry shake” on at a net rate of 3 kg/m² or pre-mixed and applied as a liquid slurry on freshly laid concrete and trowelled/power float finished in a manner as recommended by the flooring specialist. The floor hardener may comprise of cement, mineral aggregate, admixtures and pigment and must provide a heavy duty, hard wearing, abrasion resistant topping for the monolithic concrete floor.
- (e) Guarantee for Flooring Works
 - (i) The whole of the flooring works shall be executed to the satisfaction of the S.O.. The Contractor shall provide a five (5) year guarantee of the flooring system. A specimen copy of the guarantee is included in Volume 1.

G8.2.12 Waterproofing System to Internal Wet Areas

- (a) The Contractor shall apply an elastomeric seamless membrane consisting of polyurethane/bitumen compounds to form an impermeable, flexible and bonded barrier at the internal wet areas as indicated on the Schedule of Internal Building Finishes. The membrane shall bond to both the base concrete slab and the top wearing surface.
- (b) Properties

(i) Colour - ASTM D412	Black
(ii) Elongation	400 - 600%
(iii) Modulus at 100% Elongation	100 - 140 KN/sq metre
(iv) Shore A Hardness (7 days R T)	12 – 15
(v) Permeability Rating 30 mils (ASTM D1653)	0.15 perms
(vi) Permeability Rating 50 mils (ASTM D1653)	0.008 perms
(vii) Recovery (Fed Spec S.55-200-S)	85-90%
(viii) Non-Volatiles	98.4%
(ix) Tensile Strength (ASTM D2370)	340 - 480 KN/sq metre
- (c) The Contractor shall apply the elastomeric waterproofing membrane strictly in accordance with the Manufacturer's instructions. Specialist shop drawings and details shall be supplied by the Contractor for the S.O.'s approval.
- (d) The Contractor shall ensure that all substrates to receive the membrane shall be cleaned and free of water puddles, grease, dust coatings or other foreign materials.

- (e) The Contractor shall provide in the name of the Specialist Applicator/Manufacturer a Ten (10) year warranty in favour of the Board against failure of all waterproofing works carried out by the Specialist in respect of faulty workmanship and premature breakdown of materials. The warranty shall be delivered to the S.O. within 30 days of handover of the completed Works.

G8.2.13 Homogeneous Tiles

- (a) Homogeneous tiles shall be heavy duty non-slip ceramic polished/unpolished speckled finish, plain coloured homogeneous stoneware ceramic, fully vitrified manufactured to SS 301:1985. They shall be high strength and density, with low water absorption, and resistant to impact, abrasion, temperature variation, chemicals and weathering.
- (b) Homogeneous tiles shall be of one of the following sizes
- (i) Type 1 : individual tiles 200 mm x 200 mm in plan x 10 mm thick; and
 - (ii) Type 2 : individual tiles 300 mm x 300 mm in plan x 10 mm thick
- (c) The tiles shall comply with the following technical properties in accordance with (SS301:1985) or otherwise stated:
- (i) Density 2.4 gm/cm³
 - (ii) Modulus of Rupture Average Extra heavy > 38 N/mm²
 - (iii) Water Absorption in weight 0.05% maximum
 - (iv) Moh's Hardness 6 or above
 - (v) Straightness of sides Maximum 0.5%
 - (vi) Rectangularity Maximum 1.8%
 - (vii) Flatness of Surface
 - 1) Centre Curvature of Surface Shall not exceed 0.5% of the length of corresponding diagonal
 - 2) Warpage of Surface Shall not exceed 0.5% of the length of corresponding diagonal
 - 3) Edge Curvature of Surface Shall not exceed 0.5% of the nominal dimension.
 - 4) Resistance to Surface Abrasion Extra heavy duty : 130 mm
 - a) (max) (Vol of material removed mm³) Heavy duty : 160 mm³
 - b) (max) Medium heavy duty: 205 mm³
 - c) (max) Medium duty : 400 mm³ (maximum)
 - 5) Resistance to Acid Alkaline < 130 m3. No visible change.
 - 6) Colour Fastness Test No visible variation
 - 7) Colour Range Minimum of 14 distinct colours for S.O.'s selection

- (d) Before laying homogeneous tiling, the base concrete surface shall be thoroughly roughened by picking and cleaned off with stiff brushes. It shall then be wetted with water, after which one coat of cement grout shall be spread over the whole floor area ensuring that it penetrates all the small holes and crevices. While the grout is still wet, a layer of 25 mm thick cement mortar shall be spread over the surface area, compacted and smoothed.
- (e) The homogeneous tiles shall be bedded into the cement mortar and hand tamped with wooden blocks. Before the mortar sets, the grooves shall be adjusted and straightened using straight metal guides. Any excessive cement mortar on the surface area shall be cleaned with a piece of clean wet cloth. The homogeneous tiles shall be arranged in orderly symmetrical order.
- (f) The homogeneous tiling to floors shall be kept damp for a period of three days after laying and during this period it shall not be used for any traffic.

G8.2.14 Mosaic Tiles

- (a) Mosaic tiles shall be 20 mm square non-slip ceramic mosaic. The pattern and colour of the mosaic tiles shall be approved by the S.O. before the order is placed with the manufacturer or supplier. The squares shall be uniform in size and colour and shall be held in position in panels by being gummed onto a piece of paper for transport to Site.
- (b) Before laying of the mosaic tiling, the base concrete surface shall be thoroughly roughened by picking and cleaned off with stiff brushes. The concrete surface shall then be wetted with water, after which one coat of cement grout shall be spread over the whole floor area ensuring that it penetrates all the small holes and crevices. While the grout is still wet, a layer of 25 mm thick cement mortar shall be spread over the surface area and shall be compacted and smoothed.
- (c) The panels of mosaic shall be bedded into the cement mortar and shall be hand tamped with wooden blocks. As soon the gummed paper is removed by damping the surface with water, and before the mortar sets, the grooves shall be adjusted and straightened with straight metal guides. Any excessive cement mortar on the surface area shall be cleaned with a piece of clean wet cloth. The mosaic tiles shall be arranged in orderly symmetrical order.
- (d) The mosaic tiling to floors shall be kept damp for a period of three days after laying and during this period, the flooring shall not be used for any traffic.

G8.2.15 Ceramic Floor Tiles

- (a) Ceramic tiles shall be 200 mm x 200 mm x 8 mm or as specified and of the non-slip type.
- (b) Tiles shall be soaked in clean water before laying. Before laying ceramic tiling, the base concrete surface shall be thoroughly roughened by picking and cleaned off with stiff brushes. The concrete surface shall then be wetted with water, after which one coat of cement grout shall be spread over the whole floor area ensuring that it penetrates all the small holes and crevices.

- (c) While the grout is still wet, a layer of 25 mm thick cement mortar, minimum strength 15 N/mm^2 , shall be spread over the surface area and shall be compacted and smoothed before the wet tile is placed in position and tamped carefully into place. The area of tiles laid shall be limited to not more than 6 m^3 at any one time. Joints shall be filled with cement grout consisting of one part of cement and one part of sieved fine sand. All surplus grout shall be cleaned off and the floor left to set for a minimum of 4 days before being subject to non-pedestrian loading.
- (d) On completion of all work, all stains and cement residue etc., shall be cleaned off with a non-corrosive tile cleaner. The floor shall be washed thoroughly and left to dry. The tiled floor must be left clean, free from stains and to the satisfaction of the S.O..
- (e) When tiled floor finishes meet plastered walls, 100 mm skirtings of the same material as the floor tiles shall be laid on walls vertical to the floor concerned and flush with the plastered walls.

G8.2.16 Granolithic Finish

- (a) Granolithic concrete shall consist of two (2) parts of cement to five (5) parts of graded granite aggregate by volume. The granite shall be graded in accordance with Table 3 of BS 882 for all-in aggregate.
- (b) The mixing shall be as specified for concrete, and the water content shall be just sufficient to produce a dense concrete of adequate workability.
- (c) Before laying, the base concrete shall be cleaned and prepared as specified.
- (d) The granolithic finish shall be spread, screeded to allow compaction, thoroughly consolidated by tamping and floated by means of a steel trowel, screeded again if necessary, to produce an even surface true to within $\pm 0.8 \text{ mm}$ of a true plane measured in any direction.
- (e) The granolithic finish shall be laid to a thickness of 40 mm in panels not exceeding 2.5 m square and to fall where directed to an even graded surface. All panels shall be cast against permanent 25 mm x 2 mm thick aluminium dividing strips bedded on edge including fair joints when the screed length exceeds 2.5 m in both directions and between different floor finishes or as instructed on site by the S.O..
- (f) Where floor hardener and dust proofer are specified or indicated in the Schedule of Finishes, while the granolithic screed is still green, non-metallic floor hardener shall be sprinkled on at a net rate of 3 kg/m^2 and trowelled in a manner as recommended by the flooring specialist.
- (g) The floor shall be kept continuously damp for a period of not less than ten (10) days. The surfaces shall not be exposed to traffic or be worked on during this period.
- (h) When the floor is cured two coats of dust proofer shall be applied in a manner as recommended by the flooring specialist.

G8.2.17 Chemical Resistant Floor and Wall Coating

- (a) High Strength Cementitious Floor Topping
 - (i) The approved primer for high strength cementitious floor topping shall be a finely grained non-metallic powder consisting of cement, micro silica and chemical admixtures, mixed with an appropriate portion of water as per manufacturer's instruction and approved by the S.O.. The Contractor shall provide a list of similar installations where the product he is proposing has been successfully used.
 - (ii) The approved high strength mortar shall consist of a dense packing of cementitious inorganic material and shall be mixed and water to form a free-flowing mortar having high internal cohesion. The water/binder ratio shall preferably not exceed 0.26.
 - (iii) When hardened, the approved high strength mortar shall develop high early strength and possess high compressive strength and excellent wear resistance suitable for reinforced concrete pavement which is subject to heavy machinery equipment and chemical attack. The approved material will also need to be subjected to in-situ bond test. No debonding of mortar layer from the existing concrete surface will be accepted.
- (b) Physical Requirements
 - (i) The approved mortar shall also meet the following requirements:
 - 1) Maximum pore size of 2 nm (1 nmequals 1/1000 micron);
 - 2) Impermeable to diesel fuel with a maximum penetration of less than 0.5 mm, tested under DIN 1048;
 - 3) Compressive strength of 70 N/mm² at 3 days, 90 N/mm² at 7 days and 110 N/mm² at 28 days;
 - 4) Wear Resistance of 8 cm³/50 cm²;
 - 5) Modulus of Elasticity of 50,000 N/mm²; and
 - 6) Density of 2,400 kg/m³.

G8.2.18 Vinyl Sheet Flooring

- (a) Vinyl sheet flooring shall conform to BS 3261. Generally, the sheet flooring shall have qualities of quietness, resilience, resistant to heel and castor damage, durability, grease resistance and acid/alkali resistance.
- (b) Vinyl sheet shall be of an approved type which complies with the following technical properties. The colour and pattern of the sheet shall be selected by the S.O..

- (c) Floor covering shall be 2 mm thick and non-cushioned polyvinylchloride sheet flooring and tile flooring. Sheet Vinyl Flooring non-cushioned shall incorporate a pvc wear layer. Wear layer shall be homogeneous and non-layered PVC. Total construction shall be NON-ASBESTOS. Pattern and colour shall extend throughout total thickness of wear layer. Wear layer is polyurethane coated. Wear layer shall be supported by a fibreglass reinforced vinyl base. The chemical composition of the wear layer shall be minimum 99.3% plasticized and stabilized PVC and maximum 0.7% pigment.
- (i) Width : 2.0 meters (6'7")
 - (ii) Total Thickness : 2.0 mm (.080")
 - (iii) Wear Layer Thickness : 0.7 mm (.028")
 - (iv) Weight : 2.8 kg/m² (0.57 lb/sq ft)
 - (v) Roll Length : 25 m (27.3 yards)
 - (vi) Flexibility : No crack or break when bent around 6 mm diameter mandrel, according to US Federal Specification L-F-001641.
 - (vii) Residual Indentation : Meets or exceeds US Federal Specification L-F-001641; type 11. Tested to BS 3261 the residual indentation is to be less than 0.1 mm. To JISA5707 the residual indentation is to be 0.19 mm.
 - (viii) Average Total Thickness : Meets or exceeds US Federal Specification L-F-001641 for Class 1.
 - (ix) Dimensional Stability : Meets or exceeds US Federal Specification L-F 001641. Shows no measurable change. Tested to JISA5707. No change.
 - (x) Resistance to Wear : After 5,000 cycles using Taber Abraser with H22 wheels, 1,000 grams load.
Maximum Weight Loss 0.38 grams
Average Percentage Weight Loss 1%.
Average Loss in Thickness 0.125 mm.
Average Percentage Loss in Thickness 6.3%.
 - (xi) Colour Fastness : Tested to BS 1006 Smaragd 2.0 achieves the highest attainable rating (8), and to JISA5707 Class 5.

- (xii) Slip Resistance : Tested to Swedish Standard SS923515 "Flooring- determination of slip resistance" which classifies any coefficient of friction above 0.2 as safe, Smaragd 2.0 has a coefficient of above 0.4 in both wet and dry conditions. Tested to JISA5707 the slip coefficient is 0.41.
- (xiii) Recommended Load Limits : Short term load – approximately 2,900 psi (200 kg/cm²).
 Long term load - approximately 700 psi (50 kg/cm²).
- (xiv) Resistance to Rolling Loads : During 10,000 cycles of a wheel loaded approximately 360 psi, no delamination or separation etc. Residual indentation less than 0.1 mm.
- (d) The Contractor shall install sheet vinyl or tile flooring in strict accordance with the manufacturer's recommendations and installation instructions using an approved adhesive. In each case suitable adhesive (acrylic dispersion adhesive, resistant to plasticizer) has to be recommended by the adhesive producer. All seams shall be heat welded with welding rod, using approved heat welding equipment. Total installation shall be performed by an experienced qualified and approved installation contractor. All designs are to be reverse installed.
- (e) The Contractor shall submit shop drawings to show laying pattern and positions of seams for the approval of the S.O. prior to laying.
- (f) Vinyl sheet flooring shall be laid with an adhesive approved by the S.O. onto an evenly trowelled 25 mm thick cement and sand (1:3) screed.
- (g) Floor surfaces to receive vinyl sheets shall be free from dirt, sand and other particles. An approved self levelling compound shall be used where necessary to correct minor undulations in the cement and sand screed.
- (h) Vinyl skirting shall be extruded vinyl, 3 mm thick, with small splay at the bottom and fixed using approved adhesive.
- (i) The whole works shall be completed by the Contractor which includes cleaning, sealing and waxing the whole floor before handingover.

G8.2.19 Heavy Duty Rubber Tile

- (a) The resilient floor shall be Heavy Duty Rubber Tile 686 x 686 x 9.5mm thick or other approved standard size, colour to be selected by the S.O..
- (b) The resilient floor shall be supplied and installed to the approval of the S.O., and the system proposed by the Contractor shall be fit for the purpose intended. The resilient floor shall have the following Physical Properties:

Hardness - Shore A (ASTM D-2240)	60 + 5 pts
Taber Abrasion (ASTM C-501 -1kg wt. 7000 cycles)	6.0% wt. loss

Tear Resistance (ASTM D-624)	93 lbs/in + 15
Tensile Strength (ASTM D-412)	435 psi + 100
Elongation (ASTM D-412)	118% + 25
100% Modulus (ASTM D-412)	414 + 100 psi
Compression Set (ASTM D-395 method B)	12.0% + 5%
Ozone Resistance (ASTM D-1 149-50 pphm, 70 hrs.@ 104°F, Bent Loop Method)	Pass
Impact Resilience (ASTM D-2632)	31.6% Rebound
Coefficient of Friction (ASTM D-2047)	1.00 (Dry) Fire Rating
(Critical Radiant Flux (ASTM E-648 Type II)	0.25 watts/sq.cm
Horizontal Burn Rate (ASTM D-635)	15.5 mm/min
Specific Gravity	1.14 ± 0.03
Acoustic Properties	0.60 sound Absorption Coefficient
Stain Resistance	Good
Chemical Resistance	Good
Spike Traffic	Excellent
(c) The floor slab shall be thoroughly cured, dry and scaled with an approved Epoxy Sealer before installation as recommended by the manufacturer.	
(d) After preparation, the substrate shall be inspected and be approved by the S.O. before installation of the floor can proceed. All cracks, expansion joints or damaged portions of the floor shall be filled with a floor crack filler.	
(e) The Epoxy Adhesive shall be mixed and applied strictly in accordance with the manufacturer's printed instructions.	
(f) The Epoxy Adhesive shall be applied using a notched trowel meant for this purpose. It is essential that the tile be applied promptly as the adhesive sets rapidly by chemical reaction.	
(g) Tiles shall be laid S.O. as to ensure good contact with close even joints in true and continuous alignment. Tiles shall be rolled immediately after installation with a 75lb sectional roller. Excessive adhesive shall be cleared from the tile while still wet with a cloth dampened with denatured alcohol.	
(h) Edge capping strips shall be used where tile stops and the edge is exposed.	
(i) After installation, no traffic shall be allowed over the completed floor for at least 24 hours.	

G8.2.20 Carpet

- (a) All carpet shall be of the specified qualities and of design, colour and pattern selected and approved by the S.O.. Carpet shall be laid by approved Specialist complete with the necessary approved aluminium bars, grippers and other accessories for the proper installation of the carpet.

- (b) Carpet shall comply with the following technical specifications:
- | | | |
|---------|---|---|
| (i) | Construction | Woven through the back |
| (ii) | Texture | Multi-Level Loop Pile |
| (iii) | Fibre | Branded Type 6.6 Nylon BCF with Soil Resistant Finish & Built-in Static Control |
| (iv) | Yarn | 3 Ply & 2 Ply |
| (v) | Pitch | 135 |
| (vi) | Rows | 7.2 |
| (vii) | Pile Thickness | 0.50 cm |
| (viii) | Pile Weight | 1152 gm/m ² |
| (ix) | Total Weight | 2318 gm/m ² |
| (x) | Width | 3.66 m |
| (xi) | Density | 6,244 |
| (xii) | Weight Density | 212,296 |
| (xiii) | Dye Method | Yard Dyed |
| (xiv) | Static Control | Static Propensity (AATCC-134) is below normal human sensitivity of 3.5 KV or less under standard test conditions (70 F, 20% R.H.) |
| (xv) | Flammability | |
| | 1) Methenamine Pill Test
(DOC FF#1-70) | Passes |
| | 2) Flooring Radiant Panel
Test (ASTM E648) | Class 1 (exceeds 0.45 watts/sq cm Critical Radiant Flux). Direct GlueDown |
| (xvi) | Smoke Density Test | |
| (xvii) | (ASTM E662) | 450 or less |
| (xviii) | Wear Warranty | 10 Year Limited Wear Warranty |
- (c) Installation
- (i) All laying of carpet shall be executed by workpeople experienced in this work. The Contractor shall submit drawings showing positions of all seams and contain the S.O.'s approval before commencing work.
 - (ii) The Contractor shall not lay carpet until the building is weathertight, wet trades finished, the building wall dried out and all paintwork is finished and dry.
 - (iii) The laying of the carpet shall not commence until socket outlets, duct covers and other fixtures round which the carpet needs to be cut have been fixed.

- (iv) The Contractor shall fix the carpet with grippers, binder bars and trims where appropriate and form neat rolled edge at junctions with walls, columns and other vertical surfaces.
- (v) On completion, the carpet is to be well protected with dust sheets and free from dirt, sand, mud, grease, stains, etc and shall be well brushed, vacuumed, cleaned up on handing over.

G8.2.21 Raised Access Floor Systems

- (a) Where indicated on the Drawings or otherwise specified computer control rooms, SCADA room and monitoring rooms shall be provided with raised access floor systems which will permit access to the void between the elevated floor and the structural floor for the installation of cabling and pipework.
- (b) These specifications relate to performance requirements for full raised access floor systems. A system shall include the supply and installation of all components and accessories required to enable it fit for the purpose intended and to perform over the period of its designed life.
- (c) These specifications relate to the following structural grades and the Contractor may offer any type which complies with a particular grade:
 - (i) Structural Grade – Medium.
- (d) The raised access floor system (hereinafter referred to as “the floor system”) shall be 600 x 600mm removable heavy-duty modular panels including under structure and all labour, material, equipment and installation as specified.
- (e) The floor system shall consist of the complete system of load bearing fixed or removable panels supported by adjustable pedestals to provide an underfloor space for the housing and distribution of services etc. It shall include all necessary accessories, fixings and installation and floor coverings where applicable.
- (f) The floor system shall be capable of providing the specified clear cavity to accommodate electrical conduits, air delivery ducts, mechanical service lines, fire detection systems, etc.
- (g) All materials used in the floor system shall be resistant to the growth of fungi and micro-organisms and to attack by insects.
- (h) The floor system shall not contain any toxic materials or give off toxic fumes when heated which by physical contact or by inhalation of gases from such materials could prove harmful to the occupants of the building.
- (i) The top finish of the floor system shall be of appropriate high-pressure laminate. The high-pressure laminate shall not compromise the electrical resistance of 1×10^{10} Ohm of the floor system. The colour of the laminate shall be selected by the S.O..
- (j) The Contractor shall furnish a 10-year guarantee for the floor system against any defects due to design, manufacture, materials and workmanship in the standard format.
- (k) Unless otherwise specified, performance and testing of the raised access floor system shall comply fully with BS EN 12825.

- (l) The floor system shall comply with the relevant fire and safety requirements. The Contractor shall also furnish Enterprise Singapore certificate together with Contractor's PE's certification that the built system is in compliance to FSSD regulations and standards.
- (m) Structural Requirements:
 - (i) The elevated floor system and its component parts shall be designed, manufactured and installed so as to satisfy the criteria specified as follows:
 - 1) Serviceability – without deflecting such as to impair the serviceability of the floor system or part, when tested in accordance with tests specified;
 - 2) Strength – without failing when tested in accordance with tests specified; and
 - 3) The weight of each panel shall not exceed 20 kg. When submitting the system for type approval, the manufacturer is required to state the installed weight of the system per m².
 - (n) The complete floor system shall be designed for serviceability for deflection in accordance with the vertical static loads for different grades shown in Table G8.2-1. The deflection limits shall be in accordance to BS EN 12825.

Table G8.2-1: Loads for Serviceability Design for Floors

Class of Deflection (Table 2)	Class of Elements (Table 1)	Deviations on Panel Elements (Table 3)
2.5mm	≤ 10kN	Class 2

- (o) Pedestals and stringers of floor systems shall be designed for stability and strength in accordance with the loads and moments relevant to the specific grade given in Table G8.2-2.

Table G8.2-2: Loads and Moments for Stability and Strength Design for Under Structure System

Grade of floor system	Static Vertical Loads, kN		Horizontal Load, N	Horizontal Moment, Nm
	Applied over the area of pedestal head	Applied over one quadrant of pedestal head		
Medium	≤18.0	≤ 13.5	5.0	90.0

- (i) Pedestals for medium grade systems shall have a minimum base area of 9,000 mm². The bases shall be designed to transmit evenly the relevant static vertical loads given in table 4 over the full area to the sub floor. Pedestals are to be fixed to the sub floor by mechanical fixings in addition to any adhesives, unless the manufacturer can prove that without such mechanical fixings, the pedestals will not become detached from the particular sub floor, when subjected to Site Quality Control Pedestal Fixing test described in MOB Clause T42.00. Pedestals will be selected at random, at the rate of 0.5% of the number of pedestals to each area and/or floor level for this test. All pedestals are required to be fixed plumb.
- (p) Tests for Structural Compliance
 - (i) Floor panels shall be capable of sustaining the appropriate static loads and achieving deflection limits when tested in accordance with relevant test methods to MOB.
 - (ii) When so tested, the floor system shall be considered to comply with the serviceability design requirements in respect to static loading and deflection if the test specimens are capable of carrying any of the particular concentrated loads at any position with:
 - 1) A maximum deflection not exceeding 1/250 of the shortest span or 2.5mm, whichever is the least, up or down from the unloaded plane after the specimen panel is loaded for 24 h; (span may be taken as the centre to centre dimension);
 - 2) A maximum stability of 0.02mm, where stability is expressed as the difference in deflection between 23 h and 24 h after the application of the test loading;
 - 3) A residual deflection not exceeding 0.5mm when measured 2 h after the removal of the load which has been applied for 24 h; and
 - 4) A permanent deflection of the specimen panel without floor finish not exceeding 0.15mm after the removal of the test loading.
 - (iii) Floor panels shall be capable of sustaining the appropriate ultimate loads when tested in accordance to the relevant test methods specified in MOB. When so tested, the floor system shall be considered to comply with the strength design requirements in respect to ultimate loading, if the test specimens are capable of withstanding the ultimate load at any of the positions which have been subjected to the particular static load test without collapse.
 - (iv) Floor panels shall be capable of sustaining the soft body and hard body impact loads when tested to the relevant test methods in MOB. When so tested, the floor system shall be considered to comply with the dynamic impact loading requirements, if the test specimen is capable of withstanding the soft body and hard body impact load without collapse when the load is dropped from a height of 1000mm onto the following locations:
 - 1) Centre of the specimen panel; and
 - 2) Centre of one edge of the specimen panel.

- (v) Pedestals from the floor system shall be capable of sustaining the appropriate static vertical loads when tested in accordance with relevant test methods to MOB. When so tested, the pedestals shall be considered to comply with the strength requirements if the test pedestals are capable of sustaining the static vertical load for 5 min without collapse when the loading is applied over an area of 50mmx50mm at the following locations:
 - 1) Through the floor panels at the centre of the pedestal head; and
 - 2) Through the floor panels over one quadrant of the pedestal head.
- (vi) Unloaded pedestals from the floor system shall be capable of withstanding the appropriate horizontal load and moment and achieving limits in free movement at pedestal head and vertical deflection. When so tested to MOB, the pedestals shall be considered to comply with the stability requirements if the test pedestals are capable of sustaining the following:
 - 1) A horizontal load of 5 N with movement at the head of not more than 1.0mm per 100mm height; and
 - 2) A horizontal moment of 90 Nm up to a maximum applied mass of 50 kg with permanent deformation of not more than 1.0mm per 100mm in height or 0.1mm in situations where the pedestals are less than 100mm high.
- (q) The installation of the perimeters of the system and around columns etc, shall comply with all the relevant requirements of MOB.
- (r) Where a supplementary support structure is required to accommodate variation more than construction tolerances e.g. to bridge service ducts etc., it shall comply with all the relevant requirements of MOB.
- (s) The manufacturer shall state the maximum number and positions of panels that may be temporarily removed at any one time during servicing without endangering the safety of the floor system.
- (t) The structural grade of the system shall be identified by means of a colour code marking as follows:
 - (i) Medium Grade – Yellow.
- (u) The marking shall be indelibly applied to each panel, pedestal and stringer (if applicable) and shall be at least 10mm sq.
- (v) The floor system when completed, shall be sturdy, firm, rigid, free from vibration, undulations, rocking, squeaking, creep, rattles and other noises and provide a smooth and uniform finished surface and which will maintain these conditions when sufficient panels have been removed for normal access.
- (w) Floor panels shall be non-combustible, with a compliance certification to BS 476 Part 4 and a flame spread rate of Class “O” to BS 476 (Part 6 & 7) and shall be approved by Fire Safety Bureau (Singapore).
- (x) Conductivity
 - (i) The floor system shall provide a conductive path to ground to minimise the possibility of static build-up and discharge. Floor panels shall have periphery contact with pedestals for continuous electrical grounding.

- (ii) Panels shall have a maximum electrical resistance of 3 ohms or less from the top edge of the panel less surface covering to the under structure.
- (y) Detailing
 - (i) The system shall be capable of providing a close fit to all perimeters, around columns, and other such structures in both new and old buildings. It shall allow for possible movement in buildings without having any detrimental effect on the floor performance. The system shall not rely for its lateral stability on structures such as perimeter walls or columns. The maximum allowable perimeter gap of the finished floor (panel plus sealing system) shall be 2mm and edge panels shall be located to prevent movement. The method for sealing the gap shall be specified.
 - (ii) Prior to application of any load, the elevated floor shall be level to within-
 - 1) 1.5mm over any 5m square; and
 - 2) 6.0mm over any size of enclosed space.
 - (iii) The difference in height between adjacent panels without floor finish and panels with lipped edges shall not exceed 0.75mm before and after the application of any of the static loads specified.
- (z) Dimensional Tolerances
 - (i) When subjected to tests T1.00 or 22.00, the concavity or convexity of any floor panel under no load conditions shall not exceed 0.75mm when measured horizontally parallel to any edge or diagonally across a 600mm module. The deviation at any corner of a floor panel, due to twisting under no load conditions in relation to the other three corners, shall not exceed 1.00mm over a 600mm module.
 - (ii) Panel deviation of any floor panel from square shall not exceed 0.06% of the shorter diagonal length when measured to the nearest 0.01mm.
 - (iii) The deviation from the work size of the floor panel shall not exceed 0.5mm.
- (aa) Submittals
 - (i) Product Data and Manufacturer's Literature
 - 1) The Contractor shall submit the names of the types of floor system proposed, stating the manufacturer's name and country of origin to the S.O. for approval. The submission shall include manufacturer's published product data including diagrams, illustrations, and manufacturer's literature on the proposed floor system including installation instructions.
 - (ii) Shop Drawings
 - 1) The Contractor shall submit shop drawings showing the relevant details of the floor system including the following:
 - a) Floor plan layout showing grid modules, interruptions to grid, location of stringers, panels requiring drilling or cut-out for services, location of non-standard panels (if any), grilles, registers, and perforated panels;

- b) Stringer section showing material and dimensions;
 - c) Panel section showing construction, materials, dimensions and finishes;
 - d) Pedestals showing material, dimensions, limit of vertical adjustment, method of locking, methods of attachment to floor and to stringers or panels, and
 - e) Edge details and junctions with adjoining work.
- (iii) Samples
 - 1) Upon award of Contract, the Contractor shall submit a full size sample of the floor panel with finished surface and under structure components for each type of floor system to be supplied for S.O.'s approval.
 - 2) Prior to installation, the Contractor shall provide, at his own cost, 15 m2 of mock up area of each of the floor system specified for the S.O.'s approval.
- (iv) Certification and Test Reports
 - 1) All systems offered shall comply in all respects with the performance specifications in 1.4 and 1.5. Each system offered shall be tested in accordance to MOB test standards below by an approved accredited independent test laboratory. Subsequent test reports shall be submitted to S.O. to confirm that the system complies with the relevant clauses of this performance specification. Each test report shall include dimensional and material details together with a full product specification and related drawings, which shall be validated by the test authority. S.O. will require the information, as detailed in the "Schedule of Performance Offer" (see attachment Page TSP/16) with each offer.
 - a) Fire Rating (T19.00, T20.00, T40.00, T41.00);
 - b) Electrical resistance (T43.00, T44.00);
 - c) Structural (T7.00-16.00, T29.00-39.00, T42.00); and
 - d) Air leakage of air plenums (T6.00, T28.00).
- (v) Test Panels selected from Site
 - 1) Notwithstanding the test reports submitted, the S.O. shall select at random panels of the floor system from site for testing at an accredited laboratory.
- (vi) Floor Panels
 - 1) The dimensions of panels shall be 600 mm x 600 mm. All panels, with the exception of cut panels, are to be removable and fully transferable in both position and directions.

- 2) Floor panels for floor system shall be an electro-galvanised steel enclosure filled with lightweight high-strength concrete. The overall thickness of the panels shall not exceed 35mm without any surface covering. The panel perimeter shall be manufactured to receive the pedestal head location projections. Each panel shall accept a flush fit metal fastener which securely fastens each panel corner to the pedestal head.
- 3) Panels shall be protected against corrosion with a thermostatic epoxy powder coat conductive paint applied over a clean phosphate heated surface. All bare surface and edges of the panel shall be sealed with anti-dust coating.
- 4) The underside of the panels must be flat and not of hollow-out profile.
- 5) Cut-outs shall be provided on the floor panels for service outlet boxes. Cut- out size shall be as specified in the Electrical Specifications herein. Edges of cut panels shall be sealed with epoxy coating and complete with PVC protective trimming. The numbers of service outlet in the tender shall be as specified in the Electrical Specifications herein.
- 6) Floor panels shall be provided with four corrosive resistant fasteners which are retained by the pedestal heads.
- 7) The top of the floor panels shall be clad with integral high-pressure laminate finish (HPL), in compliance to approved fire-rating and flame spread. The resistance of the HPL finish shall not compromise that of the panels. The HPL shall come in a range of colours to be selected by the S.O., the final colour and finish to be able to withstand dirt and scuffing by foot traffic or wheelchairs.

(bb) Pedestal Assembly

- (i) Pedestal assembly shall be of electro-galvanised steel or adequately protected from corrosion.
- (ii) Assembly shall provide a means of levelling and locking the assembly at a selected height which requires deliberate action to change height setting. This adjusting device shall be capable of being positively locked and shall not become loose due to vibration or other causes during the life of the system.
- (iii) Assembly shall provide a minimum of 50 mm vertical adjustment, detented in 0.4 mm increments.

(cc) Pedestal Bases

- (i) Pedestal base shall be assembled to a stud or tube with fillet welds, which is designed to engage the pedestal head assembly. They shall be secured to sub floor in accordance with manufacturer's instructions. The thread stud shall be 16 mm diameter steel

(dd) Pedestal Heads

- (i) The pedestal head shall be positively locked to prevent turning. This is important to prevent loss of elevation when panels are removed. The pedestal head shall locate the panels positively and hold them in place without the need for any screw. Securing the panels by corner lock screws is not allowed.

- (ii) Pedestal heads shall be fabricated of a head plate with a corresponding stud or tube, which is designed to engage the pedestal base assembly. The head shall be designed to accept a metal fastener to mechanically lock the panels in place.
- (iii) Pedestal head shall be covered with an electrically conductive, sound-dampening pad where panels rest on pedestals.
- (ee) Stringers
 - (i) Stringers shall be of galvanised steel channel. It shall be positively positioned by bolt to pedestal head.
- (ff) Insulation
 - (i) Underfloor insulation shall be provided for all areas which are 24 hrs air-conditioned unless otherwise indicated.
 - (ii) The underfloor insulation shall be a vapour barrier insulation system consisting of double-sided aluminium foil, 25 mm thick close-cell rigid phenolic foam laminate thermal insulation with 0.4 mm thick aluminium sheet to be laid across the foam insulation to prevent mechanical damage.
 - (iii) The butt joints of the insulation boards shall be sealed with recommended bitumen emulsion seal. The joints of the aluminium foil and aluminium sheet shall be taped and fully sealed with 50 mm wide aluminium foil tape and a cove up of 100mm skirting to ensure vapour barrier.
 - (iv) The 25mm close-cell rigid thermal insulation shall comply with the following properties and specifications:

Thermal Conductivity	: 0.018
W/mK Density	: 40kg/m ³
Water absorption	: max 5%
Service temperature limits	: -180 deg to 120 deg C
Closed cell content	: 90% minimum
Fire properties	: Class 'O' to Building Code and approved by FSB (with PSB certification for BS476 parts 6 & 7)
Flammability	: Self-extinguishing
Smoke Emission	: Not more than 5% Obscuration
- (gg) Raised Access Floor Service Outlet Boxes
 - (i) The raised floor service outlet box shall be fabricated from high temperature over baked coated powder zinc alloy diecast and able to withstand 22 k N load with maximum 2.0 mm deflection. It shall come complete with hinged trap and frame and retractable lifting handles. Nylon grommet that provides protection and support of outgoing cables shall be included in each service outlet box. The service outlet box shall be suitable for 4 compartments and shall come complete with 2 Nos of 13 A twin switched socket outlets and 2 twin RJ 45 data and telephone socket outlets of TAS/NCB approved types.

- (ii) The raised floor service outlet box shall be suitable to accommodate the specified floor finish.
- (hh) Accessories
 - (i) Ramps, steps, plenum dividers, lateral bracings, fascia vertical closures, etc. of the same standard and quality to the floor system shall be provided.
 - (ii) The Contractor shall provide 2 nos. of panel lifting devices.
- (ii) Delivery, Storage and Site Handling
 - (i) The Contractor shall deliver, store, handle and protect all materials as recommended by the manufacturer from damage, moisture, corrosion, dirt and intrusion of foreign matter. Store all materials and accessories on raised and evenly levelled platforms and in a manner so as to prevent distortion, snaking or warping. Store packaged materials in the manufacturer's unopened containers, marked with the manufacturer's name and product brand name.
 - (ii) Properly palletize, crate and/or packaging for shipment and receiving at job site.
 - (iii) Handle and store on site as per written instructions of the manufacturer and protect all materials against damage, grease and other staining or disfiguring elements. Cover with an approved material during extended periods of storage.
 - (iv) Remove and dispose of all rejected and damaged goods immediately from site.
- (jj) Installation
 - (i) Floor systems and accessories shall be installed by the manufacturer's authorised representative to maintain the integrity of the products and acceptable performance of the completed installation.
 - (ii) Prior to installation, the subfloor shall be examined for unevenness, irregularities and dampness that would affect the quality and execution of the work. The Contractor shall ensure that the surface is smooth enough for the installation of raised floor.
 - (iii) Installation shall not be proceeded with until subfloor surfaces are cleaned, dried and cleared of other trades debris and ready to receive access flooring. The subfloor for the system shall be coated with a liquid plastic coating finish/anti-dust acrylic clear sealer prior to installation.
 - (iv) The floor system shall be prepared and installed in accordance with the access floor manufacturer's instructions on preparation, layout, alignment and installation.
 - (v) Installed floor system shall be level within $\pm 1.5\text{mm}$ in 3m or $\pm 2.5\text{mm}$ over the entire area. Installed floor shall be rigid and free of rocking panels.
 - (vi) On completion, the entire system shall be cleaned and protected to the satisfaction of the S.O..

(kk) Completion

- (i) Replace access floor panels which are chipped, broken, stained, scratched or otherwise damaged or rejected by the Superintending Officer resulting from the installation of the floor system or any part of the works which do not conform to specified requirements.
- (ii) Keep all completed works clean and dry and prevent unnecessary traffic. Where required, provide 3 mm thick hardboard or other approved equivalent to traffic areas and heavy-duty polythene sheeting in other areas. If heavy equipment including staging is imposed on the finished works, provide and apply additional hardboard or other approved equivalent as required.
- (iii) On completion, submit bound recommendations for the care and maintenance of the access floor, and operating instructions for panel removal and height adjustment including the following information:-
 - 1) Limitation on maximum and minimum height of floor, cross- bracing, stringers or additional pedestal fixings required above a particular height;
 - 2) Limitation on adjustability;
 - 3) Installed mass of system per square meter;
 - 4) Maximum number and positions of panels that may be temporarily removed during servicing without endangering safety of floor system;
 - 5) Method of cleaning of floor covering with particular reference to adhesive and panel substrate, wet mopping and use of waxes and floor polishes; and
 - 6) Equipotential bonding method.

(ll) Critical Spares

- (i) The Contractor shall supply floor panels (including finishes) and under structure system of each type in accordance with the following table:

Area Applied	Percentage for Maintenance	Remarks
Up to 300m ²	2%	Min 1m ²
Over 300 to 1,200m ²	1%	Min 6m ² Max 12m ²
Over 1,200 to 8,000m ²	¼%	Min 12m ²
Over 8,000 to 20,000m ²	1/8%	Min 20m ² Max 25m ²
Over 20,000m ²	1/10%	Min 25m ² Max 40m ²

- (ii) The maintenance stock shall be properly protected and upon the instruction of the S.O. passed on to the user for keeping. Certified voucher by the recipient shall be forwarded to the S.O..

(mm) Rejection

- (i) Any part of the system or component which fails to comply with the requirements of this specification in any respect whatsoever at any stage of manufacture, test, erection or on completion at site may be rejected by the S.O. either in whole or in part as he considers necessary.
- (ii) After adjustment or modification, and if so directed by the S.O., the Contractor shall submit the items for further inspection and/or tests. Any part of the access floor or components with defects of such a nature that the requirements of this Specification cannot be fulfilled by adjustment or modification shall be replaced by Contractor at his own expense.

G8.2.22 Suspended Ceilings

- (a) Unless otherwise specified, the ceiling panels shall be 20 mm thick mineral fibre boards or approved equivalent. The Contractor shall submit a sample to the S.O. for approval prior to installation.
- (b) The panels shall be suitable for use in the local climatic conditions and shall be resistant to attack by termites, pests and fungal growth.
- (c) Water Resistant Fibrous Plaster Ceiling
 - (i) Suspended fibrous plaster glass ceiling board shall be 13mm thick supported on approved suspension system (e.g. aluminium -galvanised metal) installed in accordance with the specialist/manufacturer's instructions.
 - (ii) The material shall be composed of selected grade gypsum plaster and special stranded glass fibre and shall incorporate and approved silicone based additive to increase strength and reduce porosity.
 - (iii) The water resistant fibrous plaster ceiling shall conform to the following technical data:

Plasterglass Technical Data

Density	13mm thk	1034 kg/m ³
Weight	13mm thk	13.16kg/m ²
Glass Content	13mm thk	339 gm/m ²
Strength	Flexural strength	4.8N/mm ²
Compressive strength	11.2 N/mm ²	
Thermal	Conductivity (k)	0.21W/mK
Coefficient of expansion		20x10-6/°C
Fire Performance	Surface spread of flame	Class'O'
(Building regs classification)	Ignitability	Class'O'
Smoke/toxic gas emission		Class'O'
Alkalinity	pH value	6

- (iv) All joints shall be continuously grouted with fibreglass/plaster. After grouting had set, remove all temporary railings and support. Flush up all joints and nail holes and leave surface to dry before painting.

- (d) Suspension System
 - (i) The suspension system shall be a fully concealed proprietary brand and shall comprise hangers (with thread adjustment), cold formed primary cross rails, spline rails and sliding locks all to be hot dipped galvanised as specified. The whole system shall carry twice the in-service dead loads without failure or deformations beyond the tolerances stated below:
 - 1) Ceiling departure from nominal plane: +3 mm;
 - 2) Level change of adjacent hangers at > 1200 mm c/c: +1.5 mm;
 - 3) Variation from straight by a grid member between adjacent hangers:
 - a) 0.001 of span.
 - 4) Individual hangers shall carry a load of 200 kg without failure; and
 - 5) A setting tolerance of 3 mm (non-cumulative) over 300 mm will be permitted in the horizontal plan.
- (e) Ceiling grids in place shall be set out accurately parallel to the building grid. In every room, panels shall be set out symmetrically in both directions with or without a margin at the periphery of 1/2 panel or greater as directed by the S.O..
- (f) Installation and maintenance of the ceiling panels shall be carried out in accordance with the manufacturer's recommendations and in cases where installation in typical conditions do not apply, the Contractor shall approach the manufacturer for further recommendations and installation instructions.
- (g) Openings, etc in the ceiling, if required, shall be trimmed to receive recessed light fittings and neatly cored for vent and service pipes to exact size.
- (h) The ceiling grid shall be erected to the heights shown on the Drawings and to a true and level plane.
- (i) Ceiling panels shall be installed after all wet work is completed and dried out and all building services in the ceiling space are installed.
- (j) Suspension systems shall be rigid horizontally and vertically and must allow for fine adjustments each way. All members shall be aligned for true, level surface straight line.
- (k) The building shall be examined before beginning work to determine that it is properly enclosed and the structure is in proper condition to receive panels and suspension system.
- (l) The contractor shall provide all galvanised pressed steel channel, hanger rod trimmer, brackets, bolts, etc., approval. Hardwood timber framing is not allowed to be used.
- (m) The Contractor shall provide a five (5) year warranty against sag and warpage issued in the name of the Board.
- (n) Roofing screeds shall be formed with Grade 20 mass concrete finished neatly to the fall and dimensions shown on the Drawings.

- (o) For structure not receiving a waterproof system, before commencing to screed, the roof surface shall be thoroughly roughened by picking or wire brushed to expose a fresh surface and cleaned. The concrete shall then be soaked with water over night, after which the excess water shall be removed. A thin layer of cement grout shall, just before the screed is laid, be well brushed into the roughened surface. The screed shall be formed with Grade C20 mass concrete finished neatly to the fall and dimensions shown on the Drawings. The screeding shall be kept damp for a period of seven (7) days after laying.

G8.2.23 Waterproofing System for Concrete Roofs

- (a) Waterproof Roof treatment work shall be installed where specified and on concrete roofs with a slope of less than or equal to 1:100. The waterproofing shall be carried out by an approved roofing specialist sub-contractor. The Contractor shall submit the name of the roofing specialist whom he intends to engage with his Tender. The Contractor shall submit details of the proposed waterproofing system at time of tender.
- (b) The Contractor shall provide a ten (10) year guarantee of the proprietary waterproof system against leakage due to faulty material and workmanship.
- (c) The application of the approved waterproofing system shall fully comply with the Manufacturer's instructions.

G8.2.24 Waterproofing Membrane System

- (a) Waterproofing membrane system shall be a cold Liquid Applied Elastic Seamless Membrane consisting of a polyurethane modified bitumen for horizontal application to the roofs and skirting. The compound shall be unadulterated to be used and applied directly from its sealed containers all in accordance with the manufacturer's printed instructions.
- (b) The Contractor may propose alternative waterproof membrane systems with the relevant technical information, test reports from ENTERPRISE SINGAPORE or other approved testing institutes and other additional information as required by the S.O..
 - (i) The membrane physical properties shall be as follows:

1) Colour	Black
2) Stability	Passes (6 mos. @ temps. not exceeding 80°F)
3) Hardness	75-80 (min. instantaneous reading after 14 days cure)
4) Non-Volatiles	98 + 2%
5) Tensile Strength	300 psi - 310 psi (ASTM D412-75 Die C Dumb)
6) Elongation	700% - 900 % (ASTM D412-75 Die C Dumb)
7) Recovery	85 - 90% (Fed Spec S-55-200S)
8) Non-Volatile	98 + 2 %
9) Service Temperature Range	-40°F to +200°F

- | | |
|-----------------------------------|--|
| 10) Tear Resistance | 10.5N/mm - 11N/mm |
| 11) Moisture Vapor Transmission | 0.021 - 0.129 (ASTM E96 - 66gm/hr/m ²) |
| 12) Water Vapor Permeability | 0.29 - 0.36 (ASTM E96 - 66ng/Pa.s.m) |
| 13) Water Vapor Permeance | 17 – 20 (ASTM E96- 66ng/Pa.S.m ²) |
| 14) Crack Bridging | 1.75mm - 3.0mm (ASTM C 957) |
| 15) Adhesion Strength to Concrete | 2.65N/mm - 2.7N/mm (ASTM D 836) |
- (c) Prior to application of the waterproofing membranes system, the entire surfaces shall be in a sound, and thoroughly clean and dry condition free of debris, grease, oil, cement laitance, etc. and to the satisfaction of the S.O.. Surfaces contaminated with oil, grease, water puddles and foreign materials shall be vigorously scrubbed with a power broom and kept dry. All expansion joints shall be protected by a bond breaker tape applied over the joint extending 75mm on either sides. All substrate to receive the waterproofing membrane shall be sound, clean, dry and free of debris, grease, oil.
- (d) Application of Waterproofing Treatment
- (i) The waterproofing system shall be applied in accordance with the manufacturer's recommendations. In the absence of manufacturers guidelines the sequence shall be as follows:
 - (ii) The Cold Liquid Applied Unreinforced Elastic Seamless Membrane shall be applied on top of the roof slab to a minimum thickness of 50 mils (30 sq. ft. per gallon). Extra coating shall be applied at cracks or points of stress all in accordance with the manufacturer's printed instruction.
 - (iii) A protective proprietary slush coat comprised of cement slurry mixed with bonding agent shall be laid on top of the installed membrane for protection followed by 50 mm thick waterproofed cement and sand rendering (1:3) with the addition of waterproofing admixture & laid to fall.
 - (iv) During the progress of work, the SO may at his discretion require samples of various components of the waterproof membrane being used on site for testing. All work carried out which fails to meet the prescribed standards shall be removed and the work shall be re-executed at the Contractor's own costs and expenses to the satisfaction of the S.O..
- (e) Roof Details
- (i) Detailing at eaves shall be as indicated in the Drawings. Where the waterproof membrane meets walls or roof parapets, plinths etc., the membrane shall be turned up a minimum of 300 mm high unless otherwise indicated on the Drawings.. The internal angles shall be provided with a cement and sand angle fillet
 - (ii) All reinforced concrete roof slabs, gutters, troughs, skirtings, etc. shall be provided with waterproof cement and sand screed and waterproofing membrane unless otherwise stated. Where gutters are not provided on flat roofs, the Contractor shall allow for scupper drains in order to channel all rainwater to floor outlets.

- (iii) All pipe penetrations above roof shall be fixed with approved neoprene sleeves in accordance with the manufacturer's instructions and all penetrations shall be made watertight.
- (f) Bituminous Stains
 - (i) In the event of the bituminous stains occurring on the external painted walls of the building blocks, the Contractor shall remove the stain, make good and re-paint the affected surfaces to the satisfaction of the S.O..
- (g) Tests
 - (i) The waterproofing membrane shall be visually examined by the S.O. and shall be flood tested.
 - (ii) Waterproofed areas shall be flood tested for 72 hours after the system has cured and prior to installation of protective course. Plug drains and barriers shall be placed to contain the water to 100mm depth.
 - (iii) The deck shall be flooded to a depth of 100mm for the duration of the test prior to installation of the cement panel roofing.
- (h) Flat Roof Insulation
 - (i) A minimum thickness of 50 mm Extruded Polystyrene Insulation Foam shall be applied as a thermal insulation system. The Extruded Polystyrene Insulation Foam shall be installed in strict accordance with the manufacturer's printed instructions with particular attention paid to dishing into gutters, skirting, aprons, etc and to be installed by the Specialist Contractor.
 - (ii) Physical Properties of Extruded Polystyrene Insulation

Foam: Typical Properties	Test Standard	Value
1) Material Thickness	50mm	
2) Density	DIN 53420	32 kg/m ³ ASTM D-1622 2.2lbs/ft ³
3) Thermal Conductivity	ASTM C-518	0.032W/mk
4)	ASTM C-177	0.220 Btu.in/ft ² .hr.°F
5) Compressive Strength	DIN 53421 ASTM D-1621	250 Kpa (minimum) 36 psi
6) Water Vapor	ASTM C-355	0.79 perm. cm
7)	Permeability	0.47 perm. in.
8) Water Absorption	DIN 53428	0.20% by Vol.
9)	By Submersion	ASTM D-2842 >1.00% Vol.
10) Fire Classification	DIN 4102	B1 (difficult to ignite)

- (i) Waterproof Reinforced Cement Sand Panel Roofing
 - (i) All flat roofs and roofs to fall shall be provided with 1000 x 1000 x 50 mm thick waterproofed cement and sand (1:3) mesh reinforced panels with the addition of waterproof liquid admixture agent at the rate of 850 cc per 50 kg of cement mix used in accordance with the manufacturer's printed instructions and laid over 50mm thick dense polystyrene insulation with a 100% polypropylene, UV stabilized, non-woven type geofabric separation between the insulation and the panel. Joints shall be filled with molten bitumen or polysulphide joint sealant to be approved by the S.O..
 - (ii) The waterproof cement and sand plain face shall be applied in two or more coats and finished to a smooth surface with a steel trowel.
 - (iii) Waterproofing compound used shall be suitable for the purpose and shall be approved by the S.O..

G8.2.25 Vapour Barrier

- (a) Where specified or shown on the Drawings, the Contractor shall provide a vapour barrier underlay to concrete in direct contact with the ground.
- (b) The underlay shall consist of a 5-layer membrane, made up of 2 polyethylene and Kraft laminates bonded with bitumen and a 3-way reinforced glass fibre mesh. The underlay shall have a nominal thickness of approximately 400 microns.
- (c) Water vapour flow resistance shall be not less than 90 MN s/g when tested in accordance with ASTM E96-E.
- (d) The Contractor shall lay the vapour barrier underlay on properly prepared granular fill. For tied slabs, the Contractor shall push the vapour barrier onto the starter bars against the vertical face and lap over the vapour barrier underlay already placed on granular fill.
 - (i) The Contractor shall form joints with 150 mm laps and seal with 58 mm wide PVC pressure sensitive tape. Surfaces to be jointed shall be clean and dry.
 - (ii) The Contractor shall cross slit for penetrations such as pipes and columns so that the vapour barrier underlay fits tightly around the penetration. The connection shall be taped up.
 - (iii) Concrete shall not be poured over damaged underlay. Repairs shall be made by taping down a substantial patch over the affected part.
 - (iv) Boxing shall not be held in place with pegs which penetrate the underlay. Screeds may be set over underlay on mounds of stiff concrete.
- (e) External Bituminous Waterproofing Coating to Basements and other Structures
 - (i) The concrete surface exposed to the earth for all basements and any other specified structures shall be provided with two coats of bituminous rubber emulsion coating system unless specified otherwise. The waterproofing system shall be approved by the S.O.. Concrete surfaces to receive waterproofing coating shall be of Class F2 finish, smooth, plane, clean and dry. New concrete shall not be placed against the painted surface until the paint is dry.

- (ii) The coating shall be mixed and applied in accordance with the manufacturers guidelines and recommendations.
- (iii) A protection board shall be provided to protect against physical damage to the waterproofing. The protection board shall be semi-rigid, minimum 3.5mm thick or as recommended by the waterproofing manufacturer.

G8.2.26 Pre-Applied External Waterproofing Membrane

- (a) Where specified, a pre-applied waterproofing membrane shall be provided to the external face of the concrete structure to prevent ground water ingress. The membrane shall be provided below the finished ground level only unless specified otherwise.
- (b) The membrane shall be a pre-applied HDPE waterproofing membrane such as Fosroc Proofex Engage or similar approved which can mechanically bond to the poured concrete.
- (c) Where the membrane is to be installed horizontally, to the underside of a concrete slab, the membrane shall be applied to a smooth blinding free from loose aggregate or other sharp protrusions. Standing water shall be removed to prevent contamination of overlaps.
- (d) Where installed vertically, the mechanically bonded membranes shall be installed prior to pouring concrete, within the formwork.
- (e) The membrane shall be installed in accordance with the manufacturers recommendations and the Drawings and Specification. Care shall be taken to prevent objects from piercing the membrane. Intricate details such as pipe entries, penetrations etc shall be sealed with suitable product approved by the waterproofing manufacturer.
- (f) On completion of all building finishing works all laitance and splashes shall be scraped and cleaned off floors and walls and all surfaces left clean to the satisfaction of the S.O..

G8.2.27 Partitions

- (a) Gypsum board partitions shall be of 'Fabwall' slimline demountable partitions or approved equivalent. Full height partitions shall be covered with selected '3M Scotchgard Tapetex' linen wall covering or approved equivalent on both sides.
 - (i) Gypsum boards of half-glazed partitions shall be similarly covered;
 - (ii) All partitions shall be complete with all necessary aluminium door frames, heads, trimmings & skirtings in anodised finish; and
 - (iii) All full height partitions of individual offices including doors shall be sound insulated with approved heavy density fibreglass blankets.
- (b) Drywall partition unless otherwise indicated shall consist of 12.5 mm thick non-combustible gypsum board on both sides fixed galvanized steel channel studs with non-rusting concealed self-tapping screws and infilled with mineral wool of minimum density of 60 kg/m³. The overall thickness of the partition shall be approximately 100mm thick. All gypsum boards shall come with bevelled edge.

- (c) For external facing partition, the outer partition board shall be of 12 mm thick calcium silicate board.
- (d) All partition shall be vertical laid with abutting edges occurring over studs flanges. The tapered butt joints shall be filled with ready mix joint compound, reinforced with joint tape. Screws heads and other depressions shall also be filled with joint compound. Joint compound shall be applied in 3 coats, feathered and finish surface sand smoothen to receive wall finishes as specified.
- (e) At all corners where drywall partition meets other walls such as concrete wall, brick wall, etc and at corner joint, perforated metal angle bead reinforcement fixed with fasteners at 300 mm centres, on each side shall be provided as shown in the drawing.
- (f) Expansion joints shall be provided at intervals of not more than 7.2 m with plaster sheet stopping angles and plastic or metal casing beads
- (g) Horizontal studs bracing shall be installed wherever vertical studs are cut or the gypsum board is cut for passage of pipes, ducts or other penetrations and anchor horizontal bracing to vertical bracing with sheet metal screws. All door opening shall have double vertical rows of steel studs.
- (h) Additional galvanized steel studs and stiffeners shall be provided for magnetic white board to be mounted to the drywall partition.
- (i) All partition perimeter shall be sealed with an approved acoustical sealant. All openings through the gypsum board wall shall be sealed using appropriate materials to the satisfaction of the S.O..
- (j) Gypsum wall board shall be of an approved quality to receive wall covering and paint. Steel stud framing shall be 20 Gauge Type 358 ST10 with 0.95 mm thickness with lipped edge sections, 20 gauge galvanized and screwable or other equal and approved. It shall be fixed at vertical centres not exceeding 300 mm.
 - (i) The steel studs shall be self-splicing with wide flanges knurled for easy screw penetration and shall be hot dip galvanised.
 - (ii) The gypsum boards shall be mounted onto the steel stud framing with non-rusting self-tapping screws or other equal and approved.
 - (iii) All door openings shall have a double vertical rows of steel studs.
 - (iv) The Contractor shall co-ordinate the partition installation with the installation of building services. The programme and mode of partition installation shall be planned in such a way that building services are installed and tested on sections of the partitions which have been completed on one side prior to erection of the other side of the partitions.
 - (v) The types of drywall flush partitions to be constructed shall be 16 mm AC@ gypsum panels or other equal and approved both sides where greater widths within the partition cores are required.
 - (vi) The Contractor shall ensure that all fixings required on the wall shall be fixed on to the steel studs and shall be responsible for any damage caused to the drywall flush partitions by wall fixtures not bolted to the steel studs.

- (k) Steel Studs for Chase Wall Construction
 - (i) Where greater core widths are required for pipe enclosures and other service pipes, chase walls shall be erected.
 - (ii) The chase walls shall consist of a double row of steel studs with gypsum panel cross braces between rows. Gypsum board shall be fixed to the external face. As an alternative 64 mm steel studs cross brace screw attached to the chase wall studs may be used. When chase wall studs are not directly opposite, steel stud cross brace 600 mm centres shall be anchored to the continuous horizontal 64 mm runners screw attached to the chase wall studs.
 - (iii) The maximum thickness of chase walls shall be 600 mm, with a maximum vertical brace spacing of 1200 mm centre to centre.
 - (iv) Steel studs and runners shall be fixed together with 10 mm pan head screw or other equal and approved.
- (l) Mineral Fibre Insulation
 - (i) The Contractor shall provide mineral fibre board, a lightweight thermal insulation designed as an infill in cavity walls and partitions to reduce heat flow and sound transmission on the designated areas as specified in the Schedule of Finishes. The mineral fibre board shall consist of long, fine fibres spun from molten natural rock bonded with a thermosetting resin.
 - (ii) Mineral fibre shall comply with the following technical properties:
 - 1) Nominal Density : 40 kg/m³
 - 2) Thickness : 25 mm thick
 - 3) Thermal Conductivity : 0.036 W/mk mt 20°C
 - 4) Thermal Resistance : 0.7 m²K/W
 - 5) Service Temperature : 200 degree C
 - 6) Early Fire Hazard Indices :

Ingitability (0-20)	0
Spread of Flame (0-10)	0
Heat Evolved (0-10)	0
Smoke Developed (0-10)	0
 - 7) Moisture Resistant : 0.270 Moisture absorption exposure to an atmosphere of 50°C and 95% relative humidity
- (m) The partitions including rockwool shall be installed in accordance with the manufacturer's recommendations and shall be subject to the approval of the S.O..

G8.2.28 Resin Flooring

- (a) Base

Concrete surface prepared in accordance with manufacturer's instructions.

- (b) Construction
Fully Bonded (where structural movement joints are incorporated in the base they should be reflected through the asphalt topping at the same width and to accurate alignment).
- (c) Manufacturer
Submit to the S.O. for approval.
- (d) Mix
Refer to manufacturer for specification and mix appropriate for use.
- (e) Thickness
Between 0.1-12mm (refer to manufacturers recommendations for appropriate thickness).
- (f) Finish
Trowel applied heavy duty floor, self-smoothing, or sealant all applied in accordance with manufacturers recommendations.

END OF SECTION